

THREE-DIMENSIONAL QUANTUM GRAVITY FROM MODULAR KOSZUL DUALITY

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ABSTRACT. Three-dimensional Einstein gravity with negative cosmological constant is $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern–Simons (Achucarro–Townsend, Witten). Its boundary symmetry algebra is the Virasoro algebra at central charge $c = 3\ell/(2G)$ (Brown–Henneaux). We prove that the modular Koszul duality of the Virasoro algebra (the bar complex $\bar{B}^{\text{ord}}(\text{Vir}_c)$ with its ordered deconcatenation coproduct on $\mathbb{C} \times \mathbb{R}$, the derived chiral centre $\mathcal{Z}_{\text{ch}}^{\text{der}}(\text{Vir}_c)$ carrying E_3^{top} structure via Sugawara topologisation, and the universal Maurer–Cartan element Θ_{Vir_c} whose genus expansion is the perturbative partition function) constitutes the complete algebraic description of the gravitational theory.

The shadow obstruction tower reads as the gravitational perturbation series: $\kappa(\text{Vir}_c) = c/2$ is the tree-level central charge, the quartic contact $10/[c(5c+22)]$ is the four-graviton amplitude, and the infinite depth (class **M**) means infinitely many quantum corrections. Koszul duality $\text{Vir}_c \leftrightarrow \text{Vir}_{26-c}$ is gravitational S-duality; self-duality at $c = 13$. An algebraic avatar of the Page curve emerges from complementarity: the two Koszul-dual saddles exchange dominance at Page time $t_P = 3S_{\text{BH}}/13$, c -independent because $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13$; the crossing structure is a structural shadow of the Page transition, while the full information-theoretic content requires the island mechanism (Penington, AEMM). BTZ black holes are Maurer–Cartan elements. De Sitter entropy from analytic continuation $\kappa \rightarrow -|\kappa|$.

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I. THREE-DIMENSIONAL GRAVITY AS CHERN–SIMONS

Three-dimensional Einstein gravity with negative cosmological constant $\Lambda = -1/\ell^2$ has no propagating degrees of freedom. The Riemann tensor is determined algebraically by the Ricci tensor, which equals $\Lambda g_{\mu\nu}$; every solution is locally AdS_3 . The local triviality has an exact formulation: pure gravity in $2+1$ dimensions is a gauge theory.

I.1. The Chern–Simons formulation.

Theorem 1.1 (Achúcarro–Townsend [1], Witten [2]). *Three-dimensional gravity with cosmological constant $\Lambda = -1/\ell^2$ is classically equivalent to $SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$ Chern–Simons theory at levels $k_{\pm} = \ell/(4G)$, where G is Newton’s constant. The gauge fields are*

$$A_{\pm}^a = \omega^a \pm \frac{e^a}{\ell}, \quad (1)$$

where e^a is the dreibein and ω^a the spin connection. Flatness $F_{\pm} = dA_{\pm} + A_{\pm} \wedge A_{\pm} = 0$ is equivalent to constant negative curvature plus torsion-freedom.

Proof. The Einstein–Hilbert action with $\Lambda < 0$ in three dimensions reads

$$I_{\text{EH}} = \frac{1}{16\pi G} \int \epsilon_{abc} \left(e^a \wedge R^{bc} + \frac{1}{3\ell^2} e^a \wedge e^b \wedge e^c \right),$$

where $R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}$ is the curvature two-form. The Chern–Simons functional for a gauge field A on a three-manifold is $I_{\text{CS}}[A] = (k/4\pi) \int \text{tr}(A \wedge dA + \frac{2}{3} A \wedge A \wedge A)$. Using the $\mathfrak{sl}_2$ trace $\text{tr}(J_a J_b) = \frac{1}{2} \eta_{ab}$ and the decomposition (1):

$$I_{\text{CS}}[A_+] - I_{\text{CS}}[A_-] = \frac{k}{4\pi} \int \epsilon_{abc} \left(2 e^a \wedge R^{bc} + \frac{2}{3\ell^2} e^a \wedge e^b \wedge e^c \right) = \frac{k}{2\pi} I_{\text{EH}},$$

provided $k = \ell/(4G)$. The equations of motion $F_{\pm} = 0$ decompose into the torsion constraint $T^a = de^a + \epsilon^a{}_{bc} \omega^b \wedge e^c = 0$ and the Einstein equation $R^{ab} + (1/\ell^2) e^a \wedge e^b = 0$. $\square$

Remark 1.2 (Scope). The equivalence holds at the classical level: both theories have the same equations of motion and the same moduli of flat connections. At the quantum level, the Chern–Simons path integral is a topological invariant (the Reshetikhin–Turaev functor), while the gravity path integral must also sum over topologies. The relation between these two quantisations is the subject of Sections 4–5.

1.2. The Brown–Henneaux central charge. The boundary dynamics is not trivial. Imposing asymptotically AdS_3 boundary conditions at the conformal boundary $\partial(\text{AdS}_3) \cong \mathbb{R} \times S^1$ produces a two-dimensional conformal field theory whose symmetry algebra is generated by the asymptotic charges.

Theorem 1.3 (Brown–Henneaux [3]). *The asymptotic symmetry algebra of AdS_3 gravity with $\Lambda = -1/\ell^2$ is two copies of the Virasoro algebra at central charge*

$$c = \frac{3\ell}{2G},$$

equivalently $c = 6k$ with $k = \ell/(4G)$ the Chern–Simons level.

Proof. In the Chern–Simons formulation, the boundary condition on the half-space $\mathbb{R}_{\geq 0} \times \mathbb{C}$ is the Fefferman–Graham gauge $A_z|_{t=0} = J_- + T(z) J_+$, where $J_{\pm}, J_0$ are the $\mathfrak{sl}_2$ generators with $[J_0, J_{\pm}] = \pm J_{\pm}$, $[J_+, J_-] = 2J_0$. The residual gauge transformations preserving this form are $g = \exp(\epsilon(z) J_+)$, acting by

$$\partial_{\epsilon} T = \epsilon' T + 2\epsilon T' + \frac{c}{12} \epsilon''',$$

which is the Virasoro coadjoint action. The Chern–Simons symplectic form induces the Poisson bracket

$$\{T(z), T(w)\}_{\text{PB}} = \frac{c}{2} \delta'''(z-w) + 2T(w) \delta'(z-w) + T'(w) \delta(z-w),$$

which is the defining relation of Vir_c . The central charge is $c = 6k = 3\ell/(2G)$. $\square$

1.3. The input datum. The entire construction of this paper flows from the Virasoro λ -bracket, which encodes the OPE of the boundary stress tensor in algebraic form:

$$\{T_{\lambda} T\} = \partial T + 2T\lambda + \frac{c}{12} \lambda^3. \quad (2)$$

The three terms encode, respectively: the translation $T_{(0)}T = \partial T$ (simple pole), the dilatation $T_{(1)}T = 2T$ (double pole), and the central extension $T_{(3)}T = c/2$ (quartic pole). In the OPE formalism:

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \quad (3)$$

The triple pole $T_{(2)}T = 0$ vanishes: the coefficient of λ^2 in the λ -bracket is zero, forced by skew-symmetry. The quartic pole is the single datum from which the entire gravitational structure follows.

Remark 1.4 (Physical parameters). The central charge $c = 3\ell/(2G)$ is the only free parameter of 3d gravity with $\Lambda < 0$. In Planck units $\ell = 1$, it is $c = 3/(2G)$: the inverse coupling constant. The semiclassical limit $c \rightarrow \infty$ is $G \rightarrow 0$. The curvature $\kappa(\text{Vir}_c) = c/2$ records the same information; the relation $c = 2\kappa$ persists throughout.

1.4. The holomorphic-topological structure. The Chern–Simons theory on $\mathbb{R}_{\geq 0} \times \mathbb{C}$ has a natural factorisation into holomorphic and topological directions: the $\mathbb{C}$ -direction carries the holomorphic OPE structure (the vertex algebra of the boundary), while the $\mathbb{R}_{\geq 0}$ -direction carries the topological ordering (the associative structure of the bar complex). The product $\mathbb{C} \times \mathbb{R}$ is the habitat of the E_1 -chiral bar complex $\tilde{B}^{\text{ord}}(\text{Vir}_c)$, which is a dg coalgebra over the Koszul dual cooperad $(\text{ChirAss})^1$: the bar differential uses the holomorphic OPE data from $\text{FM}_k(\mathbb{C})$, while the deconcatenation coproduct uses the ordered topological structure from $\text{Conf}_k^<(\mathbb{R})$.

The $\text{SC}^{\text{ch},\text{top}}$ structure on the chiral derived centre $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$ (see Section 3) becomes E_3 -topological via Sugawara topologisation at non-critical level: the Sugawara stress tensor $T(z)$ satisfies $T = \{Q_{\text{CS}}, G'\}$ in the BRST complex, so $\mathbb{C}$ -translations are Q -exact, the complex structure on $\mathbb{C}$ becomes irrelevant in cohomology, and the two $\text{SC}^{\text{ch},\text{top}}$ colours collapse. The result is an E_3 -topological algebra: $E_2^{\text{top}} \times E_1^{\text{top}}$ via Dunn additivity, independent of the complex structure on $\mathbb{C}$. This is proved for the affine Kac–Moody algebra $V_k(\mathfrak{sl}_2)$ at non-critical level $k \neq -2$; for the Virasoro algebra obtained by Drinfeld–Sokolov reduction, the topologisation is cohomological and the chain-level structure is conditional on A_∞ coherence.

2. THE VIRASORO BAR COMPLEX

The ordered bar complex $\tilde{B}^{\text{ord}}(\text{Vir}_c) = T^c(s^{-1}\overline{\text{Vir}_c})$ is an E_1 -chiral coassociative coalgebra: the bar differential $d_{\tilde{B}}$ comes from the OPE residues at the boundary strata of $\text{FM}_k(\mathbb{C})$, and the deconcatenation coproduct $\Delta: \tilde{B} \rightarrow \tilde{B} \otimes \tilde{B}$ is the cofree tensor coalgebra structure on $T^c(s^{-1}\overline{\text{Vir}_c})$. The bar complex is a single-coloured E_1 coalgebra; it is not an $\text{SC}^{\text{ch},\text{top}}$ -coalgebra (the Swiss-cheese structure emerges on the derived centre pair $(C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c), \text{Vir}_c)$, not on $\tilde{B}$).

2.1. The A_∞ operations. The λ -bracket (2) determines the binary A_∞ operation m_2 directly. The quartic pole forces m_2 to be non-associative; the Stasheff identity then forces all higher operations $m_3, m_4, \dots$ in perpetuity: the tower is infinite, and Vir_c is class **M**.

Definition 2.1 (The binary operation). The binary A_∞ operation for Vir_c is the λ -bracket itself:

$$m_2(T, T; \lambda) = \{T_\lambda T\} = \frac{c}{12} \lambda^3 + 2T \lambda + \partial T. \quad (4)$$

Proposition 2.2 (The Virasoro associator). *The associator $A_3 = m_2 \circ (\text{id} \otimes m_2) - m_2 \circ (m_2 \otimes \text{id})$ is*

$$A_3(T, T, T; \lambda_{12}, \lambda_{23}) = -\partial^2 T - (2\lambda_{12} + 3\lambda_{23}) \partial T - 2\lambda_{23}(2\lambda_{12} + \lambda_{23}) T - \frac{c}{12} \lambda_{23}^3 (2\lambda_{12} + \lambda_{23}). \quad (5)$$

In particular $A_3 \neq 0$ whenever $c \neq 0$ or the field-dependent terms are non-trivial: the λ -bracket is not associative. For affine Kac–Moody algebras (where the quartic pole vanishes), $A_3 = 0$ and m_2 is strictly associative.

Proof. The PVA Jacobi identity gives $A_3(\ell, \mu) = -\{T_\mu\{T_\ell T\}\}$. Direct computation using sesquilinearity of the λ -bracket: the inner bracket $\{T_\ell T\} = \partial T + 2T\ell + (c/12)\ell^3$ is substituted into the outer bracket using $\{T_\mu(\partial T)\} = -(\mu + \partial)\{T_\mu T\}$ (the left sesquilinearity identity) and $\{T_\mu T\} = \partial T + 2T\mu + (c/12)\mu^3$. The scalar term $\{T_\mu(c/12)\ell^3\} = 0$ (the bracket of a constant vanishes). Collecting terms yields (5). $\square$

The Stasheff identity $m_3 = -A_3$ then gives the ternary operation.

Proposition 2.3 (The ternary operation). *The ternary A_∞ operation for Vir_c is*

$$m_3(T, T, T; \lambda_{12}, \lambda_{23}) = \partial^2 T + (2\lambda_{12} + 3\lambda_{23}) \partial T + 2\lambda_{23}(2\lambda_{12} + \lambda_{23}) T + \frac{c}{12} \lambda_{23}^3 (2\lambda_{12} + \lambda_{23}). \quad (6)$$

The four terms encode: the second-order geometric correction ($\partial^2 T$), the first-order response $((2\lambda_{12} + 3\lambda_{23})\partial T)$, the stress-tensor backreaction $(2\lambda_{23}(2\lambda_{12} + \lambda_{23})T)$, and the three-graviton coupling $((c/12)\lambda_{23}^3(2\lambda_{12} + \lambda_{23}))$.

Proposition 2.4 (The quaternary operation). *The quaternary A_∞ operation, determined by the degree-4 Stasheff identity on $\text{FM}_4(\mathbb{C}) \cong K_4$ (five compositions: two from $(r, s) = (2, 3)$, three from $(r, s) = (3, 2)$), is*

$$\begin{aligned} m_4(T^4; \lambda_{12}, \lambda_{23}, \lambda_{34}) &= (\lambda_{23} + 2\lambda_{34}) \partial^2 T \\ &+ (5\lambda_{12}^2 - 4\lambda_{12}\lambda_{23} + 3\lambda_{12}\lambda_{34} - 5\lambda_{23}^2 + 10\lambda_{23}\lambda_{34} + 4\lambda_{34}^2) \partial T \\ &+ (\text{degree-3 polynomial}) \cdot T \\ &+ \frac{c}{12} Q_4(\lambda_{12}, \lambda_{23}, \lambda_{34}), \end{aligned}$$

where Q_4 is a homogeneous polynomial of degree 5. At the symmetric point $\lambda_{12} = \lambda_{23} = \lambda_{34} = \lambda$:

$$m_4(T^4; \lambda, \lambda, \lambda) = 3\lambda \partial^2 T + 13\lambda^2 \partial T + 14\lambda^3 T + \frac{7c}{12} \lambda^5. \quad (7)$$

The leading term is $\partial^2 T$ (at spectral depth 2), not $\partial^3 T$ (depth 1): the depth-1 coefficient vanishes by palindromic cancellation among the Stasheff compositions.

Proof. Solve the degree-4 Stasheff relation using the sesquilinearity of m_2 (4) and m_3 (6): the first-slot rule $m_3(\partial^j T, \dots) = (-\lambda_{12})^j m_3(T, \dots)$ and the last-slot rule $m_3(\dots, \partial^j T) = (\lambda_{23} + \partial)^j m_3(\dots, T)$ reduce every composition to a polynomial in $\lambda_{12}, \lambda_{23}, \lambda_{34}$. The symmetric-point evaluation (7) is verified by direct computation. $\square$

2.2. The classical r -matrix. The classical r -matrix is obtained from the λ -bracket by the collision residue construction: the bar kernel $d \log(z - w)$ absorbs one pole, so the r -matrix has poles one order lower than the OPE (the $d \log$ absorption principle).

Proposition 2.5 (Virasoro collision residue). *The collision residue of the Virasoro λ -bracket is*

$$r^{\text{coll}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}, \quad (8)$$

a cubic-plus-simple rational function. The OPE has a quartic pole; the $d \log$ absorption reduces the order by one. The Laplace kernel (without $d \log$ absorption) is $r^L(z) = (c/2)/z^4 + 2T/z^2 + \partial T/z$.

Remark 2.6 (Convention). Two conventions coexist in the literature. The collision residue (8) absorbs $d \log(z)$ (one pole fewer than OPE). The Laplace kernel $r^L(z)$ retains the full OPE poles. We use the collision residue throughout, following the bar-complex convention where the propagator is $d \log E(z, w)$ (weight 1 always). At $c = 0$ the scalar part of r^{coll} vanishes: the abelian limit, consistent with $\kappa(\text{Vir}_0) = 0$.

Proposition 2.7 (Yang–Baxter equation). *The CYBE*

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0$$

holds for the collision residue (8). It is the projection to the binary collision stratum of the Arnold relation $\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{13} + \omega_{13} \wedge \omega_{12} = 0$ on $\text{FM}_3(\mathbb{C})$, with $\omega_{ij} = d \log(z_i - z_j)$.

2.3. The c -linearity theorem. A structural pattern persists across all computed degrees m_2 through m_5 : the central charge enters each operation through a single scalar contact term, while all stress-tensor-dependent coefficients are c -independent. This separates the universal gravitational backreaction from the coupling-dependent scattering amplitude.

Theorem 2.8 (c -linearity). *For all $k \geq 2$, the scalar part of $m_k(T^{\otimes k}; \lambda_1, \dots, \lambda_{k-1})$ is **linear** in c :*

$$m_k|_{\text{scalar}} = \frac{c}{12} P_k(\lambda_1, \dots, \lambda_{k-1}),$$

where P_k is a homogeneous polynomial of degree $k + 1$. All T -dependent coefficients (the polynomials multiplying $\partial^j T$ for $0 \leq j \leq k - 2$) are independent of c .

Proof. The central charge enters the λ -bracket (2) as a scalar contact term $(c/12)\lambda^3$. Scalars are annihilated by the bracket in both slots: $\{1_\lambda T\} = 0 = \{T 1_\lambda\}$ (PVA vacuum axiom). Induction on the Stasheff relations: each m_k is a sum of compositions of lower m_j , and the only c -dependent input is the scalar part of m_2 . Since scalars are killed by subsequent brackets, c propagates linearly through the recursion. The degree count follows from $\deg P_2 = 3$ and the Stasheff composition raising degree by 2 at each step. $\square$

Remark 2.9 (Gravitational interpretation). The theorem states that graviton self-interaction at every degree is controlled by a single coupling: $c/12 = \kappa/6$, where $\kappa(\text{Vir}_c) = c/2$. The stress-tensor backreaction ($T, \partial T, \dots, \partial^{k-2} T$ terms) is universal, independent of the cosmological constant. The pure-graviton scattering amplitude is proportional to $c/12$: Newton’s constant enters once.

2.4. Infinite tower and class **M**.

Theorem 2.10 (Non-truncation). *The A_∞ tower of Vir_c does not truncate: $m_k \neq 0$ for all $k \geq 3$ and generic c . More precisely:*

- (i) *The quartic OPE pole forces $A_3 \neq 0$, hence $m_3 = -A_3 \neq 0$.*
- (ii) *The Stasheff identity at each subsequent degree forces $m_k \neq 0$ from the non-vanishing of the wheel integral on the associahedron K_k .*
- (iii) *At $c = 0$, the scalar terms vanish but the field-dependent terms ($\partial^j T$ coefficients) survive unchanged: the tower remains infinite even without central charge.*

*The discriminant $\Delta = 8\kappa S_4 = 8(c/2) \cdot S_4 \neq 0$ for generic c , confirming class **M**: infinite shadow depth.*

Proof. Part (i): the associator (5) contains the field-independent term $-\partial^2 T$, which is nonzero regardless of c , and the c -dependent term $-(c/12)\lambda_{23}^3(2\lambda_{12} + \lambda_{23})$, which is nonzero for $c \neq 0$.

Part (ii): the degree- k Stasheff identity is $\sum_{i+j=k+1} (-1)^{i(j-1)} m_i \circ m_j = 0$. The compositions involve the previously established nonzero operations $m_2, \dots, m_{k-1}$, and the obstruction at degree k is nonzero by the wheel integral computation: the Hamiltonian cycle on the complete graph K_k with $k - 1$ cubic vertices has nonzero residue from the quartic OPE pole.

Part (iii): by the c -linearity theorem (Theorem 2.8), setting $c = 0$ kills only the scalar contact terms. The field-dependent coefficients are c -independent and nonzero (the self-coupling $T_{(1)} T = 2T$ drives the induction). $\square$

Remark 2.11 (The Drinfeld–Sokolov mechanism). The infinite A_∞ tower is manufactured by Drinfeld–Sokolov reduction from the *finite* Kac–Moody tower. The principal DS reduction of $V_k(\mathfrak{sl}_2)$ produces Vir_c with $c = 1 - 6(k+1)^2/(k+2)$, transporting the complexity class from $\mathbf{L}$ (affine: $m_k = 0$ for $k \geq 3$) to $\mathbf{M}$ (Virasoro: all $m_k \neq 0$). The mechanism: the DS stress tensor $T^{\text{DS}} = J^f + (J^h)^2/(4(k+2)) + \frac{1}{2}\partial J^h$ contains a quadratic Sugawara term whose double Wick self-contraction generates the quartic OPE pole, absent in the original affine OPE. The manufactured quartic pole makes m_2 non-associative, which seeds the entire infinite tower.

2.5. The deconcatenation coproduct. The bar complex $\bar{B}^{\text{ord}}(\text{Vir}_c) = T^c(s^{-1}\overline{\text{Vir}_c})$ carries the deconcatenation coproduct

$$\Delta[s^{-1}a_1 | \cdots | s^{-1}a_n] = \sum_{i=0}^n [s^{-1}a_1 | \cdots | s^{-1}a_i] \otimes [s^{-1}a_{i+1} | \cdots | s^{-1}a_n],$$

which is the cofree coassociative coalgebra structure. Together with the bar differential $d_{\bar{B}}$, this makes $\bar{B}^{\text{ord}}(\text{Vir}_c)$ an E_1 dg coassociative coalgebra. The coproduct is structural (it exists for any bar complex); the gravitational content is in the differential $d_{\bar{B}}$, which encodes the A_∞ operations via the OPE residues.

The Drinfeld–Sokolov transfer theorem shows that the *spectral* coproduct Δ_z (the deformation of the deconcatenation by the spectral parameter) is primitive on Vir_c : $\Delta_z(T) = \tau_z(T) \otimes 1 + 1 \otimes T$, with no corrections at any order. The mechanism is a ghost-number obstruction: the homotopy h of the DS strong deformation retract has ghost degree -1 , and every tree in the HPL expansion of the transferred coproduct at degree $n \geq 2$ contains an uncompensated h -insertion that deposits the intermediate state in a ghost sector invisible to the projection. Gravitons do not split.

3. THE DERIVED CHIRAL CENTRE: THE GRAVITATIONAL BULK

The bar complex computes the boundary data. The bulk is the derived chiral centre $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) = C_{\text{ch}}^\bullet(\text{Vir}_c, \text{Vir}_c)$: the chiral Hochschild cochain complex of the boundary algebra, defined via the chiral endomorphism operad $\text{End}_{\text{Vir}_c}^{\text{ch}}$ with spectral parameters from $\text{FM}_k(\mathbb{C})$.

3.1. Chiral Hochschild cohomology of the Virasoro algebra.

Theorem 3.1 (Chiral Hochschild cohomology of Vir_c). *For generic c , the chiral Hochschild cohomology of Vir_c is concentrated in degrees $\{0, 1, 2\}$ with Hilbert series $P(t) = 1 + t^2$:*

$$\begin{aligned} \text{ChirHoch}^0(\text{Vir}_c) &= \mathbb{C}, \\ \text{ChirHoch}^1(\text{Vir}_c) &= 0, \\ \text{ChirHoch}^2(\text{Vir}_c) &= \mathbb{C} \cdot \Theta_c, \\ \text{ChirHoch}^n(\text{Vir}_c) &= 0 \quad \text{for } n \geq 3 \text{ and generic } c, \end{aligned}$$

where $\Theta_c \in \text{ChirHoch}^2$ is the Gel'fand–Fuchs 2-cocycle recording $\partial_c\{T_\lambda T\} = \lambda^3/12$.

Proof. The Hochschild–Serre spectral sequence for Vir_c collapses at E_2 for generic c . In degree 0: the centre of Vir_c is $\mathbb{C}$ (the Virasoro algebra is simple for generic c). In degree 1: every derivation is inner (the Virasoro algebra has $H^1(\text{Vir}_c, \text{Vir}_c) = 0$ for $c \neq 0$, since the only derivation D satisfying $D[L_m, L_n] = [DL_m, L_n] + [L_m, DL_n]$ is $D = \text{ad}(v)$ for some v). In degree 2: the Gel'fand–Fuchs computation gives $H^2 = \mathbb{C}$ for generic c , generated by the central extension cocycle $\Theta_c(L_m, L_n) = \frac{1}{12}(m^3 - m)\delta_{m+n,0}$. In degree ≥ 3 : vanishing by spectral sequence degeneration at generic c (the singular locus in the c -plane where higher cohomology appears is discrete). $\square$

Corollary 3.2 (The bulk of 3d gravity). *The derived chiral centre of Vir_c is*

$$Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) \simeq \mathbb{C}[[c]].$$

The unique bulk observable is the central charge parameter: 3d gravity has no local degrees of freedom.

Proof. By Theorem 3.1, the Hilbert series is $P(t) = 1 + t^2$. The degree-0 component $\mathbb{C}$ is the identity; the degree-2 component $\mathbb{C} \cdot \Theta_c$ is the deformation parameter c . The formal power series ring $\mathbb{C}[[c]]$ is the completed algebra of functions on the one-dimensional formal moduli space parametrised by the central charge. $\square$

3.2. The Swiss-cheese structure and E_3 -topologisation. The chiral derived centre carries a natural $\text{SC}^{\text{ch},\text{top}}$ structure: the pair $(Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), \text{Vir}_c)$ is a two-coloured datum where the bulk (closed colour) acts on the boundary (open colour). This is the algebraic encoding of the bulk-to-boundary map in $\text{AdS}_3/\text{CFT}_2$.

Theorem 3.3 (E_3 -topologisation for Virasoro). *For Vir_c obtained by Drinfeld–Sokolov reduction from $V_k(\mathfrak{sl}_2)$ at non-critical level $k \neq -2$ (the critical level $k = -h^\vee = -2$ maps to $c \rightarrow \infty$ under $c = 1 - 6(k+1)^2/(k+2)$; the condition excludes this singularity), the $\text{SC}^{\text{ch},\text{top}}$ structure on the derived centre pair promotes to an E_3 -topological structure on $H^\bullet(Z_{\text{ch}}^{\text{der}}(\text{Vir}_c), Q)$. The mechanism: the Sugawara stress tensor satisfies $T(z) = \{Q_{\text{CS}}, G'(z)\}$ in the BRST complex, so $\mathbb{C}$ -translations are Q -exact, the two $\text{SC}^{\text{ch},\text{top}}$ colours collapse in cohomology, and $E_2^{\text{hol}} \times E_1^{\text{top}} \rightarrow E_2^{\text{top}} \times E_1^{\text{top}} = E_3^{\text{top}}$ via Dunn additivity.*

Remark 3.4 (Scope). The E_3 -topological structure is proved cohomologically: it holds on $H^\bullet(Z^{\text{der}}, Q)$, not necessarily at the cochain level. For class **M** algebras (Virasoro), the chain-level E_3 is conditional on $\mathcal{A}_\infty$ coherence of the topologisation. The cohomological statement suffices for the genus-tower computations of Section 4. The proof is for affine Kac–Moody $V_k(\mathfrak{sl}_2)$ at non-critical level, transported to Virasoro by the DS intertwining theorem; general chiral algebras with conformal vector are conjectural.

3.3. The bulk-to-boundary map. The bulk-to-boundary map sends the degree-2 chiral Hochschild class $\Theta_c \in \text{ChirHoch}^2$ to the stress tensor $T \in \text{Vir}_c$ via the conformal Ward identity:

$$\beta_B(\partial_c)(z_o) \cdot T(w) = \frac{T(w)}{(z_o - w)^2} + \frac{\partial T(w)}{z_o - w}. \quad (9)$$

This completes the *Koszul triangle*: bulk $\xrightarrow{\beta_B}$ boundary $\xrightarrow{\bar{B}}$ lines $\xrightarrow{Z^{\text{der}}}$ bulk. The bar complex is the universal coalgebra from which three functors extract the three vertices:

- (i) $\Omega(\bar{B}) \simeq \text{Vir}_c$ (cobar recovers the boundary; bar-cobar adjunction),
- (ii) $\Omega(D_{\text{Ran}}(\bar{B})) \simeq \text{Vir}_{26-c}$ (Verdier dual followed by cobar gives the Koszul dual; see Section 5),
- (iii) $\text{RHom}(\Omega(\bar{B}), \text{Vir}_c) \simeq \mathbb{C}[[c]]$ (chiral Hochschild cochains give the derived centre = bulk).

4. THE SHADOW TOWER AS GRAVITATIONAL PERTURBATION SERIES

The universal Maurer–Cartan element Θ_{Vir_c} in the modular convolution algebra $\mathfrak{g}_{\text{Vir}_c}^{\text{mod}}$ satisfies the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ and encodes the perturbative expansion of 3d gravity. Its projections to successive degrees constitute the *shadow obstruction tower*, a sequence of invariants $\{S_r\}_{r \geq 2}$ that record the perturbative corrections to the semiclassical gravitational dictionary.

4.1. The shadow invariants.

Definition 4.1 (Shadow tower of Vir_c). The shadow obstruction tower is the sequence $S_2, S_3, S_4, \dots$ of projections of the MC element Θ_{Vir_c} to successive degrees in the convolution algebra. The first four named invariants are:

- (i) $S_2 = \kappa(\text{Vir}_c) = c/2$: the curvature of the bar complex (UNIFORM-WEIGHT).

- (ii) $S_3 = \mathfrak{C}(\text{Vir}_c) = -c$: the cubic shadow, encoding the Lie bracket structure of the convolution algebra.
- (iii) S_4 : the quartic shadow, which determines the quartic contact invariant

$$\mathfrak{Q}_{\text{Vir}}^{\text{contact}} = \frac{10}{c(5c + 22)}. \quad (10)$$

The numerator 10 arises from the Shapovalov inner product at level 4; the denominator $c(5c + 22)$ is the Kac determinant. Poles at $c = 0$ ($G = \infty$) and $c = -22/5$ (Lee–Yang) mark reorganisation points of the tower.

- (iv) S_r for $r \geq 5$: the higher shadow obstructions, each determined by the degree- r Stasheff identity and the Kac determinant at levels $\leq r$.

4.2. The discriminant and class **M**.

Theorem 4.2 (Discriminant). *The discriminant of Vir_c is*

$$\Delta(\text{Vir}_c) = 8\kappa S_4 = 8 \cdot \frac{c}{2} \cdot S_4 \neq 0 \quad (11)$$

for generic c (in particular for all $c > 0$). The discriminant is linear in κ , not quadratic. Since $\Delta \neq 0$, the shadow tower does not terminate: Vir_c is class **M** with infinite shadow depth $r_{\max} = \infty$.

Proof. The quartic OPE pole forces $S_4 \neq 0$ (Proposition 2.4): the scalar contact term $(7c/12)\lambda^5$ at the symmetric point is proportional to c and nonzero for $c \neq 0$. Combined with $\kappa = c/2 \neq 0$ for $c \neq 0$: $\Delta = 8(c/2)S_4 \neq 0$. When $\Delta \neq 0$, the recursion governing the shadow tower propagates to all genera, giving $r_{\max} = \infty$. $\square$

4.3. Gravitational interpretation of the shadow degrees. Each shadow degree has a direct gravitational reading. The table:

Shadow	Algebraic content	Gravitational counterpart
$\kappa = c/2$	bar complex curvature	Brown–Henneaux central charge; one-loop Weyl anomaly
$\mathfrak{C} = -c$	cubic Lie bracket	cubic graviton vertex $\propto \ell/(16\pi G)$
$\mathfrak{Q} = \frac{10}{c(5c+22)}$	quartic contact	four-graviton amplitude; genus-2 contact correction
$S_r, r \geq 5$	degree- r obstruction	r -graviton vertex; higher-loop correction

4.4. The genus tower. The curvature $\kappa = \kappa(\text{Vir}_c) = c/2$ controls the genus expansion of the gravitational partition function through the fiberwise differential $d_{\text{fib}}^2 = \kappa \cdot \omega_g$, where $\omega_g = c_1(\mathbb{E})$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$.

Theorem 4.3 (The intrinsic genus tower). *The intrinsic genus- g free energy of Vir_c is (UNIFORM-WEIGHT, all $g \geq 1$):*

$$F_g^{\text{intr}}(\text{Vir}_c) = \kappa(\text{Vir}_c) \cdot \lambda_g^{\text{FP}}, \quad \lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g = \frac{2^{2g-1} - 1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!} > 0.$$

The generating function is the $\hat{A}$ -genus:

$$\sum_{g \geq 1} F_g^{\text{intr}} x^{2g} = \frac{c}{2} (\hat{A}(ix) - 1), \quad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}.$$

The first two terms:

$$F_1^{\text{intr}} = \frac{\kappa}{24} = \frac{c}{48},$$

$$F_2^{\text{intr}} = \frac{7\kappa}{5760} = \frac{7c}{11520}.$$

Proof. The genus- g free energy is $F_g^{\text{intr}} = \kappa \cdot \int_{\overline{M}_g} \lambda_g$ by the scalar genus tower theorem applied to the single-generator algebra Vir_c (uniform weight: one generating field of conformal weight 2). The Hodge integrals λ_g^{FP} are the Faber–Pandharipande values, whose generating function is $\sum_{g \geq 1} \lambda_g^{\text{FP}} x^{2g} = \hat{A}(ix) - 1 = (x/2)/\sin(x/2) - 1$ by the Mittag-Leffler expansion $(x/2)/\sin(x/2) = 1 + 2 \sum_{g \geq 1} \frac{(2^{2g-1}-1)|B_{2g}|}{(2g)!} x^{2g}$. The substitution $x \mapsto ix$ converts $\sinh(ix/2) = i \sin(x/2)$, producing the $\hat{A}(ix)$ form with all positive coefficients (the Wick rotation from Lorentzian to Euclidean signature).

For F_1^{intr} : $\lambda_1^{\text{FP}} = (2^1 - 1)|B_2|/(2^1 \cdot 2!) = 1 \cdot (1/6)/(2 \cdot 2) = 1/24$, using $B_2 = 1/6$; then $F_1^{\text{intr}} = (c/2)(1/24) = c/48$.

For F_2^{intr} : $\lambda_2^{\text{FP}} = (2^3 - 1)|B_4|/(2^3 \cdot 4!) = 7/(8 \cdot 24) \cdot (1/30)$; since $B_4 = -1/30$, $|B_4| = 1/30$, $\lambda_2^{\text{FP}} = 7 \cdot (1/30)/(8 \cdot 24) = 7/(5760)$. Then $F_2^{\text{intr}} = (c/2)(7/5760) = 7c/11520$. $\square$

Remark 4.4 (Why the $\hat{A}$ -genus, not Todd). Three independent reasons force the $\hat{A}$ -genus: (a) *symmetry*: the genus tower uses only even powers of x ; the Todd class has odd powers; (b) *KO orientation*: the Hodge bundle $\mathbb{E}$ on $\overline{M}_g$ carries a real structure via Serre duality ($\mathbb{E}^* \simeq \mathbb{E} \otimes \omega$); real vector bundles are oriented by KO-theory, whose multiplicative genus is $\hat{A}$; (c) *explicit integration*: the Faber–Pandharipande integrals match the $\hat{A}(ix)$ coefficients through $g = 6$.

Remark 4.5 (Intrinsic versus effective). The intrinsic free energies $F_g^{\text{intr}} = \kappa \cdot \lambda_g^{\text{FP}}$ use $\kappa(\text{Vir}_c) = c/2$ and are the fundamental algebraic invariants. The *effective* free energies incorporate ghost subtraction from the bc ghost system ($c_{\text{ghost}} = -26$, $\kappa_{\text{ghost}} = -13$):

$$F_g^{\text{eff}} = \kappa_{\text{eff}} \cdot \lambda_g^{\text{FP}}, \quad \kappa_{\text{eff}} = \kappa + \kappa_{\text{ghost}} = \frac{c - 26}{2}.$$

The cancellation $\kappa_{\text{eff}} = 0$ at $c = 26$ is matter-ghost cancellation (the bosonic string critical point), not vanishing of the intrinsic curvature ($\kappa(\text{Vir}_{26}) = 13 \neq 0$). Throughout this paper, F_g without superscript denotes F_g^{intr} ; the effective version appears only in contexts involving ghost coupling.

4.5. The quartic contact correction at genus 2.

Proposition 4.6 (Genus-2 contact correction). *The quartic contact shadow (10) contributes a genus-2 correction via the $r = 4$ planted-forest graph amplitude:*

$$F_2^{\text{contact}} = \mathfrak{Q} \cdot \int_{\overline{M}_2} \psi_1^2 = \frac{10}{c(5c + 22)} \cdot \frac{1}{240} = \frac{1}{24 c(5c + 22)},$$

where $\int_{\overline{M}_2} \psi_1^2 = 1/240$ by Faber’s formula. In the semiclassical limit $c \rightarrow \infty$: $F_2^{\text{contact}} \sim 1/(120 c^2)$.

Remark 4.7 (Convergence dichotomy). The shadow tower exhibits a sharp analytic dichotomy: the *scalar sector* (contact terms $\propto c/12$) has an algebraic generating function $H(t) = t^2 \sqrt{Q_{\text{Vir}}(t)}$ with the shadow metric $Q_{\text{Vir}}(t) = c^2 + 12ct + \frac{180c+872}{5c+22} t^2$; its Taylor series converges for $|t| < R_{\text{scal}} = c \sqrt{(5c + 22)/(180c + 872)}$. The *field sector* (T -dependent terms, c -independent) diverges factorially: $\|m_k|_T\| \sim C^k \cdot k!$ (Gevrey-1), requiring Borel resummation. The Borel sum is unambiguous because the spectral curve $u^2 = Q_{\text{Vir}}(t)$ has genus 0 (no real branch points for $c > 0$): 3d gravity has no resurgent ambiguity.

5. KOSZUL DUALITY AS GRAVITATIONAL S-DUALITY

The Virasoro algebra has a Koszul dual: it is another Virasoro algebra, at the complementary central charge $26 - c$. This algebraic duality is the gravitational incarnation of S-duality; the self-dual point $c = 13$ is the fixed point of the Koszul involution.

5.1. The Koszul dual.

Theorem 5.1 (Virasoro Koszul duality). *The Koszul dual of Vir_c is Vir_{26-c} :*

$$\text{Vir}_c^! = \text{Vir}_{26-c}.$$

The Koszul complementarity constant is

$$K(\text{Vir}) = \kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = \frac{c}{2} + \frac{26-c}{2} = 13. \quad (12)$$

Self-duality holds at $c = 13$: $\text{Vir}_{13}^! = \text{Vir}_{13}$.

Proof. The bar complex $\bar{B}(\text{Vir}_c)$ is a dg coalgebra. The Ran-space Verdier dual $D_{\text{Ran}}(\bar{B}(\text{Vir}_c))$ is a dg algebra. By the bar-cobar theorem (Theorem A of Volume I), the cobar of $D_{\text{Ran}}(\bar{B})$ recovers an algebra: on the Koszul locus, this algebra is Vir_{26-c} .

The central charge shift $c \mapsto 26 - c$ arises from the Shapovalov transposition: the Koszul dual of the Verma module at conformal weight h over Vir_c is the contragredient Verma module at weight h over Vir_{26-c} . The shift by 26 is the bc ghost central charge: the Verdier duality functor on the bar complex has the same effect as coupling to the bc ghost system.

The complementarity (12) is immediate from $\kappa(\text{Vir}_c) = c/2$ and $\kappa(\text{Vir}_{26-c}) = (26 - c)/2$. $\square$

5.2. **The two critical central charges.** The Koszul involution $c \mapsto 26 - c$ has two distinguished points.

Proposition 5.2 (Two critical points).

- (i) $c = 13$: **Koszul self-duality.** $\text{Vir}_{13}^! = \text{Vir}_{13}$. *The boundary algebra equals its Koszul dual: $\kappa = \kappa' = 13/2$. The genus tower is symmetric under the involution: $F_g(c) = F_g(26 - c)$ for all $g \geq 1$. The graviton resolvent $\mathcal{G}(t; \lambda)$ acquires a $c \mapsto 26 - c$ symmetry. This is the algebraic self-duality of 3d gravity; it is not the string theory critical point.*
- (ii) $c = 26$: **matter-ghost cancellation.** *The effective curvature $\kappa_{\text{eff}} = (c - 26)/2 = 0$: the effective genus tower $F_g^{\text{eff}} = 0$ for all $g \geq 1$ (the intrinsic tower $F_g^{\text{intr}} = 13 \lambda_g^{\text{FP}} \neq 0$). The gravitational partition function becomes S-invariant (modular weight vanishes). This is the bosonic string critical point. The Koszul dual $\text{Vir}_{26}^! = \text{Vir}_0$ is the Witt algebra at $c = 0$: uncurved ($\kappa = 0$) but non-formal (infinite A_∞ tower from the field-dependent terms).*

Proof. Part (i): at $c = 13$, the involution $c \mapsto 26 - c$ fixes c , so $\text{Vir}_{13}^! = \text{Vir}_{26-13} = \text{Vir}_{13}$. The symmetry $F_g(c) = F_g(26 - c)$ follows from $F_g = (c/2)\lambda_g^{\text{FP}}$: replacing c by $26 - c$ gives $((26 - c)/2)\lambda_g^{\text{FP}}$; the two are equal when $c/2 = (26 - c)/2$, i.e. $c = 13$.

Part (ii): at $c = 26$, $\kappa_{\text{eff}} = (c - 26)/2 = 0$ directly, so $F_g^{\text{eff}} = 0$. The effective genus-1 partition function $Z_1^{\text{eff}}(\tau) = \eta(\tau)^{-\kappa_{\text{eff}}} = \eta(\tau)^0 = 1$. The Koszul dual is Vir_0 ; by Theorem 2.10(iii), Vir_0 has $\kappa = 0$ (uncurved) but $m_k \neq 0$ for all $k \geq 3$ (non-formal). $\square$

5.3. **The complementarity constant and Page time.** The complementarity constant $K = 13$ has physical content beyond its algebraic origin.

Proposition 5.3 (Complementarity and the genus tower). *For any $g \geq 1$:*

$$F_g(\text{Vir}_c) + F_g(\text{Vir}_{26-c}) = 13 \cdot \lambda_g^{\text{FP}}.$$

The sum is independent of c : the total intrinsic genus- g free energy of the Koszul pair is a topological constant equal to the complementarity constant $K = 13$ times the Hodge integral.

Proof. $F_g(\text{Vir}_c) + F_g(\text{Vir}_{26-c}) = \frac{c}{2} \lambda_g^{\text{FP}} + \frac{26-c}{2} \lambda_g^{\text{FP}} = 13 \lambda_g^{\text{FP}}$. $\square$

Remark 5.4 (Verdier duality and the modular S -transform). The Koszul involution $\mathbb{D}: c \mapsto 26 - c$ (Verdier duality on the bar complex) and the modular S -transform $S: \tau \mapsto -1/\tau$ commute on genus-1 MC elements: $S \circ \mathbb{D}(\Theta^{(1)}(\tau)) = \mathbb{D} \circ S(\Theta^{(1)}(\tau)) = \Theta_{\text{Vir}_{26-c}}^{(1)}(-1/\tau)$. They act on different variables (c vs. τ). The commutativity produces a paired Cardy formula: $S_{\text{Cardy}}(\text{Vir}_c; h) + S_{\text{Cardy}}(\text{Vir}_{26-c}; h) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{(26-c)h/6}$, which is maximised at $c = 13$ by concavity of $\sqrt{\cdot}$.

5.4. The quartic contact under Koszul duality.

Proposition 5.5 (Quartic complementarity). *Under the Koszul involution $c \mapsto 26 - c$:*

$$\mathfrak{Q}(c) + \mathfrak{Q}(26 - c) = \frac{10}{c(5c + 22)} + \frac{10}{(26 - c)(152 - 5c)}.$$

At $c = 13$ (self-dual): $\mathfrak{Q}(13) = 10/(13 \cdot 87) = 10/1131$. The quartic contact invariant is the first nonlinear witness of the duality between boundary and line sectors.

Remark 5.6 (The gauge-gravity-matter trichotomy). The complexity classification by the discriminant Δ organises the standard landscape into three columns:

	Gauge (CS/KM)	Pure gravity (Vir/ W_N)	Matter-coupled (BP)
Class	L	M	M
Shadow depth	3	∞	∞
A_∞ tower	finite	infinite	infinite
Δ	0	$\neq 0$	$\neq 0$

The passage from gauge ($\Delta = 0$, integrable) to gravity ($\Delta \neq 0$, chaotic) is Drinfeld–Sokolov reduction: the Sugawara construction manufactures the quartic pole from double poles, and **L** $\rightarrow$ **M**.

6. THE HOLOGRAPHIC DATUM $\mathcal{H}(\text{Vir}_c)$

The six preceding sections assembled the gravitational Koszul data from the single input (2). We now organise the output into the *holographic modular Koszul datum*: a six-component package that encodes the complete algebraic content of 3d gravity with negative cosmological constant.

6.1. The six-fold package.

Definition 6.1 (Holographic datum of Vir_c). The *holographic datum* of 3d gravity at central charge $c = 3\ell/(2G)$ is the sextuple

$$\mathcal{H}(\text{Vir}_c) = (\text{Vir}_c, \text{Vir}_{26-c}, C_c, r_c(z), \Theta_c, \nabla^{\text{hol}}),$$

where the six components are:

- (i) Vir_c : the **boundary algebra**, the Virasoro vertex algebra at central charge $c = 3\ell/(2G)$, with OPE (3) and $\kappa(\text{Vir}_c) = c/2$;
- (ii) Vir_{26-c} : the **Koszul dual**, the line-operator algebra (Theorem 5.1), with $\kappa(\text{Vir}_{26-c}) = (26 - c)/2$;
- (iii) C_c : the **line category**, the dg-category of perfect Vir_c -modules, carrying the braided monoidal structure induced by the r -matrix;
- (iv) $r_c(z) = (c/2)/z^3 + 2T/z$: the **classical r -matrix** (Proposition 2.5), the collision residue of the Virasoro λ -bracket with cubic-plus-simple poles;

- (v) Θ_c : the **universal Maurer–Cartan element** in the modular convolution algebra $\mathfrak{g}_{\text{Vir}_c}^{\text{mod}}$, whose projections constitute the shadow obstruction tower $\{S_r\}_{r \geq 2}$ (Definition 4.1);
- (vi) ∇^{hol} : the **shadow connection**, the flat connection on the configuration-space bundle determined by the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

Remark 6.2 (The Koszul triangle). The first three components form the *bulk-boundary-line triangle*:

$$\begin{aligned} \text{Bulk} &= Z_{\text{ch}}^{\text{der}}(\text{Vir}_c) \simeq \mathbb{C}[[c]] \\ \text{Boundary} &= \text{Vir}_c \\ \text{Lines} &= \text{Vir}_{26-c}\text{-mod (on Koszul locus)} \end{aligned}$$

The arrows are: bulk $\xrightarrow{\beta_B}$ boundary (the conformal Ward identity (9)), boundary $\xrightarrow{\bar{B}}$ lines (the bar complex followed by Verdier duality gives Vir_{26-c} -modules), lines $\xrightarrow{Z^{\text{der}}}$ bulk (chiral Hochschild cochains close the triangle).

6.2. Explicit computation of the six components. Each component of $\mathcal{H}(\text{Vir}_c)$ admits a closed-form expression.

Theorem 6.3 (The holographic datum: explicit form). *The six components of $\mathcal{H}(\text{Vir}_c)$ are determined by the single datum c :*

- (i) *Boundary curvature: $\kappa(\text{Vir}_c) = c/2$; dual curvature: $\kappa(\text{Vir}_{26-c}) = (26-c)/2$; complementarity: $\kappa + \kappa' = 13$.*
- (ii) *Shadow obstruction tower through degree 5:*

$$\begin{aligned} S_2 &= \kappa = c/2 && (\text{curvature}), \\ S_3 &= -c && (\text{cubic shadow}), \\ S_4 &= \frac{10}{c(5c+22)} && (\text{quartic contact}), \\ S_5 &= \frac{-48}{c^2(5c+22)} && (\text{quintic}). \end{aligned}$$

At large c (semiclassical): $S_4 \sim 2/c^2 \sim G^2$, $S_5 \sim -48/(5c^3) \sim G^3$.

- (iii) *Classical r -matrix: $r_c(z) = (c/2)/z^3 + 2T/z$; at $c = 0$: $r_0(z) = 2T/z$ (field-dependent only, no central extension); at $c = 13$: $r_{13}(z) = (13/2)/z^3 + 2T/z$ (self-dual).*
- (iv) *Spectral coproduct: strictly primitive, $\Delta_z(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. No higher terms at any degree: gravitons do not split.*
- (v) *Shadow connection: $\nabla^{\text{hol}} = d - \sum_{r \geq 2} S_r \omega_{(r)}$, where $\omega_{(r)}$ is the degree- r component of the bar-complex propagator form on the configuration space $\text{FM}_n(\mathbb{C})$. Flatness of ∇^{hol} is equivalent to the MC equation for Θ_c .*
- (vi) *Discriminant: $\Delta = 8\kappa S_4 = 40/(5c+22) \neq 0$ for $c \neq -22/5$. Class **M**: infinite shadow depth.*

Proof. Each component was computed in the preceding sections: (i) equation (12); (ii) Definition 4.1 for S_2 – S_4 , and the quintic shadow from the shadow-metric expansion $H(t) = t^2 \sqrt{Q_{\text{Vir}}(t)}$ with $Q_{\text{Vir}}(t) = c^2 + 12ct + (180c + 872)t^2/(5c+22)$, giving $S_5 = [t^5]H/5 = -48/[c^2(5c+22)]$; (iii) equation (8); (iv) the DS ghost-number obstruction (Section 2); (v) the MC equation of Section 4; (vi) equation (11). $\square$

6.3. The shadow connection and soft graviton hierarchy. The shadow connection ∇^{hol} decomposes the MC element into a hierarchy of Ward identities. The first three are classical; the rest are new.

Proposition 6.4 (Shadow connection and soft gravitons). *The shadow connection ∇^{hol} on $\text{FM}_n(\mathbb{C})$ acts on n -point correlation functions of primary operators $\mathcal{O}_i$ of weight h_i . Its decomposition by shadow degree gives:*

- (i) Degree 2 ($S_2 = c/2$): the Weinberg leading soft graviton theorem. Soft factor: $S_i^{(0)} = \partial_{z_i}/(z_0 - z_i)$.
- (ii) Degree 2 (double pole of S_2): the Cachazo–Strominger subleading soft theorem. Soft factor: $S_i^{(1)} = h_i/(z_0 - z_i)^2$.
- (iii) Degree 3–4: the cubic shadow $S_3 = -c$ is gauge-trivial; the first non-trivial sub-subleading contribution is the quartic contact $S_4 = 10/[c(5c + 22)]$, the Cachazo–He–Yuan soft factor requiring the ternary A_∞ operation m_3 .
- (iv) Degree $r \geq 5$: one independent Ward identity per degree, forced by the class **M** property ($m_k \neq 0$ for all $k \geq 3$). The infinite A_∞ tower produces infinitely many soft theorems beyond the known three.

At large c : the soft factors at degree r scale as $S_r \sim G^{r-2}$, and the semiclassical limit retains only S_2 (Weinberg) and the subleading $S^{(1)}$ (Cachazo–Strominger).

Proof. The shadow connection is flat (from the MC equation), so each degree- r component S_r gives an independent Ward identity. The degree-2 components extract the conformal Ward identity, whose simple-pole and double-pole residues are the Weinberg and Cachazo–Strominger soft factors. The cubic shadow is gauge-trivial: $S_3 = d_2 \sigma_3$ for an explicit gauge transformation σ_3 (using the fact that the Virasoro algebra is simple for generic c , so $H^1(\text{Vir}_c, \text{Vir}_c) = 0$). The quartic contact $S_4 = 10/[c(5c + 22)]$ is genuinely new by the non-vanishing of the Kac determinant at level 4. For degree ≥ 5 : non-vanishing follows from the infinite tower (Theorem 2.10). $\square$

6.4. The spectral braiding.

Proposition 6.5 (Virasoro fusion kernel as braiding). *The braided monoidal structure on $\mathcal{C}_c$ is determined by the r -matrix $r_c(z) = (c/2)/z^3 + 2T/z$ via the KZ-type connection on the category of Virasoro modules:*

$$\nabla_{\text{KZ}} = d - \sum_{i < j} r_c(z_i - z_j) dz_{ij}, \quad (13)$$

whose monodromy is the Virasoro fusion kernel (the Ponsot–Teschner 6 j -symbol [5]). The KZ connection has an irregular singularity at $z_i = z_j$ of Poincaré rank 2 (from the cubic pole), in contrast to the regular-singular KZ connection of affine Kac–Moody algebras (where the r -matrix has only a simple pole). The irregular singularity produces Stokes phenomena in the braiding, not present in the KM setting.

Proof. The classical r -matrix (8) determines the connection 1-form $\sum r_c(z_{ij}) dz_{ij}$ on $\text{Conf}_n(\mathbb{C})$. The CYBE (Proposition 2.7) is equivalent to flatness of this connection. The Poincaré rank at $z_i = z_j$ is the order of the pole minus one: $(z_{ij})^{-3}$ gives rank 2. The monodromy of the flat connection around $z_j = z_i + \epsilon e^{2\pi i \theta}$ produces the braiding on modules; for the Virasoro algebra, this is the Ponsot–Teschner kernel. $\square$

Remark 6.6 (Gravity versus gauge theory). The holographic datum distinguishes gravity from gauge theory through three structural signatures:

	Gauge ($V_k(\mathfrak{g})$)	Gravity (Vir_c)
r -matrix pole	simple ($1/z$)	cubic ($1/z^3$)
Coproduct	non-trivial	primitive
$\mathcal{A}_\infty$ tower	finite ($m_k = 0, k \geq 3$)	infinite (all $m_k \neq 0$)
KZ singularity	regular	irregular (Poincaré rank 2)
Shadow depth	3 (class L)	∞ (class M)
Discriminant	$\Delta = 0$	$\Delta \neq 0$

The passage from gauge to gravity is Drinfeld–Sokolov reduction: the Sugawara construction manufactures the quartic pole from double poles, simultaneously creating the infinite $\mathcal{A}_\infty$ tower, the irregular KZ singularity, and the primitive coproduct.

7. THE GENUS EXPANSION AND THE $\hat{\mathcal{A}}$ -GENUS

The shadow obstruction tower of Section 4 produces a sequence of genus- g free energies F_g through the scalar genus tower theorem. This section derives the same free energies from an independent method: the Grothendieck–Riemann–Roch theorem applied to the Hodge bundle on $\overline{\mathcal{M}}_g$. The agreement of the two derivations is a non-trivial consistency check.

7.1. The genus-1 free energy and the Weinberg soft theorem.

Theorem 7.1 (Genus-1 free energy). *The genus-1 free energy of Vir_c is (UNIFORM-WEIGHT):*

$$F_1(\text{Vir}_c) = \frac{\kappa(\text{Vir}_c)}{24} = \frac{c}{48}.$$

Three independent derivations:

- (i) **Shadow tower.** $F_1 = \kappa \cdot \lambda_1^{\text{FP}}$, where $\lambda_1^{\text{FP}} = \int_{\overline{\mathcal{M}}_{1,1}} \lambda_1 = 1/24$ and $\kappa(\text{Vir}_c) = c/2$.
- (ii) **Dedekind eta function.** The genus-1 partition function is $Z_1(\tau) = \text{Tr}_{\text{Vir}_c}(q^{L_0 - c/24}) = q^{-c/24} / \prod_{n \geq 1} (1 - q^n) = q^{-c/24} \eta(\tau)^{-1}$, where $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ is the Dedekind eta function. The free energy $F_1 = -\log |Z_1|^2|_{q \rightarrow 0} = c/48 \cdot \log |q|^{-2} = c/48$ (leading term).
- (iii) **Soft graviton.** The Weinberg leading soft factor $S_i^{(0)} = \partial_{z_i} / (z_0 - z_i)$ applied to the genus-1 one-point function gives $\langle T \rangle_{g=1} = -c/24$ (the Casimir energy on the torus). The free energy $F_1 = \kappa \cdot \lambda_1^{\text{FP}} = (c/2)(1/24) = c/48$.

Proof. Derivation (i) follows from the genus tower (Theorem 4.3) at $g = 1$, using the intrinsic curvature $\kappa = c/2$ (not the effective curvature).

Derivation (ii): the vacuum character of Vir_c for generic c (no null vectors) is $\chi_0(\tau) = q^{-c/24} \prod_{n \geq 1} (1 - q^n)^{-1}$. The full partition function on a genus-1 surface with modular parameter τ includes the full Fock space: $Z_1(\tau) = q^{-c/24} \prod_{n \geq 1} (1 - q^n)^{-1}$. The leading q -expansion gives the free energy $F_1 = c/48$.

Derivation (iii): the genus-1 one-point function is

$$\langle T \rangle_{g=1} = \frac{\text{Tr}(L_0 q^{L_0 - c/24})}{\text{Tr}(q^{L_0 - c/24})} = q \partial_q \log Z_1(\tau) = -\frac{c}{24} + O(q).$$

This is the Casimir energy density. The integrated free energy is $F_1 = (c/2) \cdot (1/24)$. □

7.2. The genus-2 free energy.

Theorem 7.2 (Genus-2 free energy). *The genus-2 free energy is (UNIFORM-WEIGHT):*

$$F_2(\text{Vir}_c) = \frac{7\kappa(\text{Vir}_c)}{5760} = \frac{7c}{11520}.$$

The Hodge integral value $\lambda_2^{\text{FP}} = \int_{\overline{\mathcal{M}}_2} \lambda_2 = 7/5760$ uses $B_4 = -1/30$ and $(2^3 - 1)/2^3 = 7/8$.

Proof. From the genus tower (Theorem 4.3): $F_2 = \kappa \cdot \lambda_2^{\text{FP}} = (c/2)(7/5760) = 7c/11520$. The computation of λ_2^{FP} : $\lambda_2^{\text{FP}} = (2^{2 \cdot 2 - 1} - 1)|B_{2,2}|/(2^{2 \cdot 2 - 1} \cdot (2 \cdot 2)!) = (2^3 - 1)|B_4|/(2^3 \cdot 4!) = 7 \cdot (1/30)/(8 \cdot 24) = 7/5760$. $\square$

Remark 7.3 (The quartic contact correction). The shadow free energy $F_2 = 7\kappa/5760$ is the scalar-lane contribution. The quartic contact shadow adds a correction (Proposition 4.6): $F_2^{\text{contact}} = \mathfrak{Q} \cdot \int_{\overline{\mathcal{M}}_2} \psi_1^2 = 10/[c(5c + 22)] \cdot 1/240 = 1/[24c(5c + 22)]$. At large c : $F_2^{\text{contact}} \sim 1/(120c^2)$, which is $O(G^2)$, a genuine two-loop quantum gravity correction invisible in the one-loop $\hat{A}$ -genus formula.

7.3. The GRR derivation. The $\hat{A}$ -genus formula for the genus tower admits an independent derivation from the Grothendieck–Riemann–Roch theorem, providing a second proof that does not depend on the shadow obstruction tower.

Theorem 7.4 (GRR derivation of the genus tower). *Let $\pi: C_g \rightarrow \overline{\mathcal{M}}_g$ be the universal curve, and $\mathbb{E} = \pi_* \omega_{C_g/\overline{\mathcal{M}}_g}$ the Hodge bundle. The Grothendieck–Riemann–Roch theorem gives:*

$$\text{ch}(\mathbb{E}) = \pi_*(\text{td}(\omega_{C_g/\overline{\mathcal{M}}_g})) = g + \sum_{k \geq 1} \frac{B_{2k}}{(2k)!} \kappa_{2k-1},$$

where κ_{2k-1} is the Miller–Morita–Mumford class and B_{2k} are Bernoulli numbers. The generating function of the Hodge integrals $\lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g$ is determined by the $\hat{A}$ -genus:

$$\sum_{g \geq 1} \lambda_g^{\text{FP}} x^{2g} = \hat{A}(ix) - 1, \quad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}.$$

Proof. The Chern character of the Hodge bundle is computed by GRR: $\text{ch}(\mathbb{E}) = \pi_*(\text{td}(\Omega_\pi^1))$, where td is the Todd class. The top Chern class $\lambda_g = c_g(\mathbb{E})$ is extracted from $\text{ch}(\mathbb{E})$ via the Newton identities relating Chern characters to Chern classes. The generating function of $\lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_g} \lambda_g$ is then determined by integrating the Todd class over the fibres of π : the fibre integral produces the $\hat{A}$ -genus of the tangent bundle of C_g (which has real rank 2, so the KO orientation produces $\hat{A}$ rather than Todd). $\square$

Remark 7.5 (Why $\hat{A}$, not Todd: three reasons). The GRR derivation confirms Remark 4.4:

- (a) *Even powers*: the genus tower uses x^{2g} (even); the Todd class has odd powers.
- (b) *KO orientation*: the Hodge bundle carries a real structure from Serre duality $\mathbb{E}^* \simeq \mathbb{E} \otimes \omega$; the real vector bundle is oriented by KO-theory, whose multiplicative genus is $\hat{A}$.
- (c) *Explicit verification*: the Faber–Pandharipande integrals match $\hat{A}(ix)$ through $g = 6$.

7.4. Uniform-weight structure for the Virasoro algebra.

Proposition 7.6 (Uniform-weight and absence of cross-channel corrections). *The Virasoro algebra Vir_c has a single generating field $T(z)$ of conformal weight 2. The genus- g free energy $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) receives no cross-channel correction: $\delta F_g^{\text{cross}} = 0$ for all $g \geq 1$.*

Proof. The cross-channel correction $\delta F_g^{\text{cross}}$ arises from the interaction between fields of different conformal weights in the propagator sum. The Virasoro algebra has a single generator of weight 2: there is only one channel, and the cross-channel correction vanishes identically. The scalar formula $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ is exact at all genera for this single-weight algebra. $\square$

7.5. The genus-1 fibre differential. The explicit genus-1 bar complex provides a direct computation of F_1 . The genus-1 propagator on the elliptic curve E_τ is $\eta^{(1)}(z_1, z_2; \tau) = \partial_{z_1} \log \vartheta_1(z_1 - z_2 | \tau) + 2\pi i \operatorname{Im}(z_1 - z_2) / \operatorname{Im} \tau$.

Proposition 7.7 (The three-term formula). *The fibre differential on the degree-2 bar element $[s^{-1}T | s^{-1}T]$ at genus 1 is:*

$$d_{\text{fib}}^{(1)}[s^{-1}T | s^{-1}T] = s^{-1} \left(:TT: - \frac{2\pi^2}{3} E_2(\tau) \cdot T - \frac{c\pi^4}{90} E_4(\tau) \cdot \mathbf{1} \right),$$

where $E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n$ is the quasi-modular Eisenstein series and $E_4(\tau)$ is the weight-4 Eisenstein series. The three terms arise from:

- (i) $:TT:$: the genus-0 normal-ordered product (regular term paired with $1/u$);
- (ii) $-\frac{2\pi^2}{3} E_2(\tau) \cdot T$: the genus-1 state-dependent correction (double pole $2T/u^2$ paired with the propagator coefficient $-(\pi^2/3)E_2(\tau)u$);
- (iii) $-\frac{c\pi^4}{90} E_4(\tau) \cdot \mathbf{1}$: the genus-1 vacuum correction (quartic pole $(c/2)/u^4$ paired with $-(\pi^4/45)E_4(\tau)u^3$).

Proof. The meromorphic part of the genus-1 propagator has the Laurent expansion $h(u; \tau) = 1/u - (\pi^2/3)E_2(\tau)u - (\pi^4/45)E_4(\tau)u^3 - \dots$ (odd powers only). The fibre differential is the residue of $h(u; \tau) \cdot T(z_1) T(z_2)$ at $u = 0$. The Virasoro OPE (3) has poles at orders 4, 2, and 1 in $u = z_1 - z_2$. Pairing the propagator terms $c_k(\tau) u^{2k-1}$ with the OPE terms $T_{(2k-1)} T u^{-(2k)}$ gives the residue: $T_{(-1)} T = :TT:$ from the leading $1/u$, $c_1 \cdot T_{(1)} T = -(\pi^2/3)E_2 \cdot 2T$ from $k = 1$, $c_2 \cdot T_{(3)} T = -(\pi^4/45)E_4 \cdot (c/2)$ from $k = 2$. $\square$

The E_2 -correction is quasi-modular (it transforms anomalously under $\tau \mapsto -1/\tau$); the E_4 -correction is fully modular. The Virasoro OPE pole structure generates precisely the Eisenstein tower $\{E_{2k}\}$ at genus 1.

8. BTZ BLACK HOLES AS MAURER-CARTAN ELEMENTS

The BTZ black hole of Bañados, Teitelboim, and Zanelli [4] is the unique black hole solution of 2+1-dimensional gravity with $\Lambda < 0$. We identify it as a Maurer-Cartan element of the gravitational convolution algebra.

8.1. The BTZ solution. A BTZ black hole is specified by two parameters: mass M and angular momentum J , subject to $|J| \leq M\ell$. In the Chern-Simons formulation, the BTZ solution is a pair of flat $\mathfrak{sl}_2$ -connections with holonomy around the spatial circle:

Definition 8.1 (BTZ gauge configuration). The BTZ black hole with mass M and angular momentum J is the flat connection $A_\pm$ with holonomy eigenvalues

$$\exp(\pm 2\pi\sqrt{h}) \quad \text{and} \quad \exp(\pm 2\pi\sqrt{\bar{h}}),$$

where the conformal weights are

$$h = \frac{M\ell + J}{2}, \quad \bar{h} = \frac{M\ell - J}{2}.$$

The outer horizon radius is $r_+ = \ell\sqrt{8GM}(1 + \sqrt{1 - (J/(M\ell))^2})^{1/2}$.

8.2. BTZ as MC element.

Theorem 8.2 (BTZ as Maurer–Cartan element). *The BTZ black hole of mass M and angular momentum J is a Maurer–Cartan element*

$$\alpha_{\text{BTZ}} = h e_h + \bar{h} e_{\bar{h}} \in \mathfrak{g}_{\text{Vir}_c}^{\text{mod}},$$

where $e_h, e_{\bar{h}}$ are the generators of the modular convolution algebra at conformal weights $h = (M\ell + J)/2$, $\bar{h} = (M\ell - J)/2$. The MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ is equivalent to the flatness condition $F_{\pm} = 0$ for the BTZ gauge connection.

Proof. The BTZ solution is a flat $SL(2, \mathbb{R})$ connection on the solid torus $D \times S^1$ with specified holonomy. In the boundary Virasoro description, flatness becomes the MC equation: the highest-weight state $|h, \bar{h}\rangle$ in the Virasoro module of weight $(h, \bar{h})$ satisfies the conformal Ward identities, which are exactly the components of the MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ projected to the boundary. $\square$

8.3. The Cardy entropy. The statistical entropy of the BTZ black hole is computed from the asymptotic growth of Virasoro characters.

Theorem 8.3 (Cardy formula and Bekenstein–Hawking entropy). *The microstate entropy of the BTZ black hole at conformal weights $(h, \bar{h})$ with $h, \bar{h} \gg c$ is:*

$$S_{\text{Cardy}}(h, \bar{h}) = 2\pi\sqrt{\frac{c h}{6}} + 2\pi\sqrt{\frac{c \bar{h}}{6}} = S_{\text{BH}},$$

where the Bekenstein–Hawking entropy is

$$S_{\text{BH}} = \frac{\pi r_+}{2G} = \frac{\text{Area}(\text{horizon})}{4G}.$$

Proof. The Cardy formula: the number of Virasoro primaries of weight h grows as $\rho(h) \sim \exp(2\pi\sqrt{ch}/6)$ for $h \gg c$ (the saddle-point approximation to the modular S -transform of the partition function). For the BTZ state $(h, \bar{h})$: $S = \log \rho(h) + \log \rho(\bar{h}) = 2\pi\sqrt{ch}/6 + 2\pi\sqrt{c\bar{h}}/6$.

The Bekenstein–Hawking entropy in terms of boundary data: using $c = 3\ell/(2G)$, $h = (M\ell + J)/2$, $\bar{h} = (M\ell - J)/2$:

$$\begin{aligned} S_{\text{Cardy}} &= 2\pi\sqrt{\frac{3\ell}{12G}(M\ell + J)} + 2\pi\sqrt{\frac{3\ell}{12G}(M\ell - J)} \\ &= 2\pi\sqrt{\frac{\ell}{4G} \cdot \frac{M\ell + J}{1}} + 2\pi\sqrt{\frac{\ell}{4G} \cdot \frac{M\ell - J}{1}} \\ &= \frac{\pi r_+}{2G}, \end{aligned}$$

where the last step uses the relation $r_{\pm}^2 = 8G\ell^2 M(1 \pm \sqrt{1 - (J/(M\ell))^2})/2$ for the non-rotating case $J = 0$: $r_+ = \ell\sqrt{8GM}$, $S_{\text{BH}} = \pi\ell\sqrt{8GM}/(2G) = \pi r_+/(2G)$. $\square$

8.4. Quantum corrections to BTZ entropy. The shadow obstruction tower provides a systematic genus expansion of quantum corrections to the leading Bekenstein–Hawking entropy.

Proposition 8.4 (BTZ genus expansion). *The entropy of the BTZ black hole admits a saddle-point expansion:*

$$S(M) = S_{\text{BH}} - \frac{3}{2} \log S_{\text{BH}} + \sum_{g=2}^{\infty} \frac{C_g(\kappa)}{S_{\text{BH}}^{2g-2}},$$

where $S_{\text{BH}} = 2\pi\sqrt{ch/6}$ and the coefficients $C_g(\kappa)$ depend on the shadow free energies $F_1, \dots, F_g$:

- (i) The leading correction is logarithmic: $-\frac{3}{2} \log S_{\text{BH}}$, universal across all 2+1-dimensional black holes.
- (ii) $F_1 = \kappa/24 = c/48$: the one-loop free energy from the scalar shadow.
- (iii) $F_2 = 7\kappa/5760 = 7c/11520$: the two-loop free energy.
- (iv) The shadow partition function $Z^{\text{sh}} = \exp(\sum_{g \geq 1} F_g \beta^{2g})$ converges absolutely (Bernoulli decay $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$), in contrast to the string free energy which diverges as $(2g)!$.

Proof. The saddle-point expansion of $Z(\beta) = \exp(\sum_{g \geq 0} F_g \beta^{2g})$ around the BTZ saddle gives the $1/S_{\text{BH}}$ expansion. The logarithmic correction $-3/2$ is the universal contribution from the one-loop determinant of the graviton fluctuation around the BTZ geometry (the Jacobian of the saddle-point integral in three dimensions). The higher-order corrections C_g are determined by the shadow free energies through the saddle-point formula. $\square$

8.5. BTZ complementarity and the Hawking–Page transition.

Proposition 8.5 (BTZ paired entropy). *The Koszul pair $(\text{Vir}_c, \text{Vir}_{26-c})$ induces a paired entropy:*

$$S_{\text{BH}}(\text{Vir}_c; h) + S_{\text{BH}}(\text{Vir}_{26-c}; h) = 2\pi\sqrt{\frac{h}{6}}(\sqrt{c} + \sqrt{26-c}),$$

maximised at $c = 13$ by the concavity of $\sqrt{\cdot}$, where $\sqrt{c} + \sqrt{26-c}|_{c=13} = 2\sqrt{13}$. The Hawking–Page transition at $\beta_{\text{HP}} = 2\pi\ell$ exchanges the BTZ and thermal AdS saddles.

Proof. The paired entropy is $S_{\text{BH}}(\text{Vir}_c) + S_{\text{BH}}(\text{Vir}_{26-c}) = 2\pi\sqrt{ch/6} + 2\pi\sqrt{(26-c)h/6} = 2\pi\sqrt{h/6}(\sqrt{c} + \sqrt{26-c})$. By AM–QM inequality, $\sqrt{c} + \sqrt{26-c}$ is maximised when $c = 26 - c$, i.e. $c = 13$. The Hawking–Page temperature $\beta_{\text{HP}} = 2\pi\ell$ is the point where the BTZ free energy $F_{\text{BTZ}} = -(c/6\pi)(\pi^2/\beta)$ crosses the thermal AdS free energy $F_{\text{AdS}} = -(c/6\pi)\beta$. $\square$

Remark 8.6 (The entanglement entropy). The Calabrese–Cardy entanglement entropy of a boundary interval of length L is $S_{\text{EE}} = (c/3) \log(L/\varepsilon)$. The Koszul-dual entanglement entropy is $S'_{\text{EE}} = ((26-c)/3) \log(L/\varepsilon)$. Their sum: $S_{\text{EE}} + S'_{\text{EE}} = (26/3) \log(L/\varepsilon)$, independent of c . At $c = 13$: $S_{\text{EE}} = S'_{\text{EE}} = (13/3) \log(L/\varepsilon)$.

9. THE PAGE CURVE FROM COMPLEMENTARITY

The complementarity constant $K = \kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13$ controls the transition between the two Koszul-dual phases of black hole evaporation. The derivation below does not use the Calabrese–Cardy entanglement rate as external input: it constructs the radiation entropy directly from the Lagrangian decomposition of the genus expansion evaluated on the BTZ background, derives the evaporation rate from the genus-1 free energy, and identifies the Page time as the crossing of the two saddle-point contributions. The c -independent Page time $t_P = 3S_{\text{BH}}/13$ is a consequence of the complementarity constant, not an assumption.

9.1. The Lagrangian decomposition on the BTZ background. The complementarity theorem (Theorem 5.1) decomposes the genus- g bar complex into Koszul-dual halves. We evaluate this decomposition on the BTZ Maurer–Cartan element $\alpha_{\text{BTZ}} \in \mathfrak{g}^{\text{mod}}$ (Theorem 8.2), producing a two-saddle partition function parametrised by the black hole mass.

Construction 9.1 (Lorentzian BTZ genus expansion). Let α_{BTZ} be the MC element at conformal weight h (non-rotating, $J = 0$). The genus-summed free energy (Proposition 8.4) is:

$$\mathcal{F}(h; \cdot) = S_{\text{BH}}(h) \cdot^{-2} + \sum_{g \geq 1} F_g(\alpha_{\text{BTZ}}) \cdot^{2g-2}, \quad (14)$$

where $S_{\text{BH}}(h) = 2\pi\sqrt{ch/6}$ and $\tau = S_{\text{BH}}^{-1}$. The Lorentzian continuation replaces the Euclidean modular parameter $\tau = i\beta/(2\pi\ell)$ by $\tau \mapsto t/(2\pi\ell) + i\epsilon$, where t is Lorentzian time and $\epsilon \rightarrow 0^+$ selects the retarded contour. For an evaporating black hole, the conformal weight is time-dependent: $h(t) = h_0 - \dot{h}t$ with $\dot{h} > 0$, so that $S_{\text{BH}}(t) = 2\pi\sqrt{c h(t)/6}$ decreases monotonically.

Theorem 9.2 (Lagrangian decomposition on the BTZ background). *The complementarity decomposition (Theorem 5.1, Theorem C) applied to the genus expansion (14) produces:*

$$\mathcal{F}(h; \beta) = \mathcal{F}_+(\kappa; h; \beta) + \mathcal{F}_-(\kappa'; h; \beta),$$

where the two Lagrangian contributions are:

$$\begin{aligned} \mathcal{F}_+(\kappa; h; \beta) &= \kappa^{-2} + \sum_{g \geq 1} \kappa \lambda_g^{\text{FP}} \kappa^{-2g-2}, \quad \kappa = c/2, \\ \mathcal{F}_-(\kappa'; h; \beta) &= \kappa'^{-2} + \sum_{g \geq 1} \kappa' \lambda_g^{\text{FP}} \kappa'^{-2g-2}, \quad \kappa' = (26-c)/2. \end{aligned}$$

The genus- g obstruction decomposes:

$$\mathbf{C}_g(\text{Vir}_c)|_{\alpha_{\text{BTZ}}} = \mathbf{Q}_g^+(\kappa) \oplus \mathbf{Q}_g^-(\kappa'),$$

with fiberwise differentials satisfying $(d_{\text{fib}}^+)^2 = \kappa \cdot \omega_g$ and $(d_{\text{fib}}^-)^2 = \kappa' \cdot \omega_g$ on the two summands.

Proof. Theorem C applied to the Virasoro algebra gives the Lagrangian eigenspace decomposition of the genus- g bar complex under the Koszul involution $c \mapsto 26-c$. The involution has period 2; its two eigenvalues on the fiberwise differential $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ are $\kappa = c/2$ and $\kappa' = (26-c)/2$, the curvatures of the two eigenspaces. Since $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT, Proposition 7.6), each summand contributes proportionally to its curvature eigenvalue. The genus-summed partition function decomposes accordingly. The evaluation on α_{BTZ} preserves the decomposition: the MC equation $D\alpha + \frac{1}{2}[\alpha, \alpha] = 0$ is satisfied independently on each Lagrangian half because the Koszul involution commutes with the MC operator (it acts on the parameter c , not on the MC variable α). $\square$

9.2. The radiation entropy from saddle-point competition. The two Lagrangian halves compete as the black hole evaporates. The radiation entropy is determined by the saddle-point prescription (the gravitational path integral selects the geometry minimising the generalised entropy), and the Page curve emerges from the exchange of dominance.

Definition 9.3 (Evaporation rate from the genus expansion). The entanglement production rate of a 2d CFT at central charge c and Hawking temperature T_H is determined by the genus-1 free energy. In thermal time units $\beta/(2\pi)$:

$$\dot{S}_{\text{rad}} = \frac{4\pi F_1}{\beta/(2\pi)} = 4\pi \cdot \frac{c}{48} = \frac{\pi c}{12} = \frac{c}{6} \text{ per unit } \frac{\beta}{2\pi},$$

using $F_1 = \kappa/24 = c/48$ (Theorem 7.1). The Koszul dual boundary, at central charge $26-c$, has entanglement production rate $(26-c)/6$.

Theorem 9.4 (The gravitational Page curve). *The radiation entropy of an evaporating BTZ black hole is*

$$S_{\text{rad}}(t) = \min(S_+(t), S_{\text{BH}}(0) - S_-(t)),$$

where:

- $S_+(t) = (c/6)t$ is the no-island contribution (entanglement growth at the boundary rate $\dot{S}_+ = c/6$, controlled by $\kappa = c/2$);

- $S_-(t) = ((26 - c)/6) t$ is the island contribution (entanglement growth at the dual rate $\dot{S}_- = (26 - c)/6$, controlled by $\kappa' = (26 - c)/2$);
- $S_{\text{BH}}(o)$ is the initial Bekenstein–Hawking entropy.

The following are consequences:

- (i) **Page time.** The crossing condition $S_+(t_P) = S_{\text{BH}}(o) - S_-(t_P)$ gives:

$$t_P = \frac{3 S_{\text{BH}}(o)}{K} = \frac{3 S_{\text{BH}}(o)}{13}, \quad (15)$$

independent of c .

- (ii) **Page entropy.** $S_{\text{Page}} = S_+(t_P) = \frac{c S_{\text{BH}}(o)}{26}$.

- (iii) **Hawking phase** ($t < t_P$): the boundary saddle $\bar{B}(\text{Vir}_c)$ with curvature $\kappa = c/2$ dominates; radiation entropy grows at rate $c/6$.

- (iv) **Island phase** ($t > t_P$): the Koszul dual saddle $\bar{B}(\text{Vir}_{26-c})$ with $\kappa' = (26 - c)/2$ dominates; radiation entropy decreases.

- (v) **Self-dual point** ($c = 13$): $\kappa = \kappa' = 13/2$, the two saddles have equal weight at all times, $S_{\text{Page}} = S_{\text{BH}}/2$.

Proof. The proof has three steps: the Lagrangian decomposition of the replica partition function, the identification of the radiation entropy from saddle dominance, and the computation of the crossing time.

Step 1: The replica partition function. The radiation entropy is computed from the replica partition function $Z(n)$ on the n -sheeted BTZ geometry: $S_{\text{rad}} = \lim_{n \rightarrow 1} (1 - n)^{-1} \log(Z(n)/Z(1)^n)$. The Lagrangian decomposition (Theorem 9.2) gives $Z(n) = Z_+(n) + Z_-(n)$, corresponding to the boundary and dual saddles. The n -sheeted replica of the BTZ geometry is itself a BTZ geometry with conical excess at the replica branch point; Theorem 8.2 identifies it as an MC element, so the genus expansion retains the same two-saddle decomposition at every replica number.

Step 2: The min formula from the generalised entropy. In the semiclassical regime $S_{\text{BH}} \gg 1$, the gravitational path integral is dominated by the saddle with smallest generalised entropy (the quantum extremal surface prescription). Two saddles contribute:

- The *no-island saddle*: the boundary-side partition function Z_+ , contributing $S_{\text{rad}} = S_+(t)$ to the radiation entropy. The rate $\dot{S}_+ = c/6$ is the genus-1 entanglement rate (Definition 9.3).
- The *island saddle*: the Koszul dual partition function Z_- , contributing $S_{\text{rad}} = S_{\text{BH}}(o) - S_-(t)$. The subtraction arises because the island saddle computes the complement: the radiation entropy equals the total entropy minus the interior entropy, and the interior is controlled by $\kappa' = (26 - c)/2$.

The dominant saddle is whichever gives the *smaller* radiation entropy: $S_{\text{rad}} = \min(S_+, S_{\text{BH}}(o) - S_-)$.

Step 3: The crossing time. The crossing condition $S_+(t_P) = S_{\text{BH}}(o) - S_-(t_P)$ reads:

$$\frac{c}{6} t_P = S_{\text{BH}}(o) - \frac{26 - c}{6} t_P.$$

Adding the two rates: $t_P(\frac{c}{6} + \frac{26 - c}{6}) = t_P \cdot \frac{26}{6} = S_{\text{BH}}(o)$. Solving:

$$t_P = \frac{6 S_{\text{BH}}(o)}{26} = \frac{3 S_{\text{BH}}(o)}{13} = \frac{3 S_{\text{BH}}(o)}{K}.$$

The c -independence is exact: the rates $c/6$ and $(26 - c)/6$ are individually c -dependent, but their sum $26/6 = \varkappa/3$ depends only on the scalar complementarity invariant $\varkappa := \kappa + \kappa' = \mathfrak{I}_3$ (distinct from the Trinity ghost-charge conductor $K(\text{Vir}) := c + c' = 26$; related by $\varkappa = \rho_{\text{Vir}} \cdot K$, $\rho_{\text{Vir}} = 1/2$). $\square$

Remark 9.5 (The entanglement rate is not an external input). The Calabrese–Cardy entanglement rate $c/6$ is not imported from an independent argument. It is *derived* from the bar complex: the genus-1 free energy $F_1 = \kappa/24 = c/48$ (Theorem 7.1) determines the Hawking radiation rate via $\dot{S}_{\text{rad}} = 4\pi F_1/(\beta/(2\pi)) = c/6$ (Definition 9.3). The rate for the Koszul dual is $(26 - c)/6$, and their sum $26/6 = K/3$ is the quantity controlling the Page time. The bar complex provides both the two saddles and their relative growth rates; the entire Page curve is internal to the genus expansion.

Remark 9.6 (c -independence of the Page time). The coefficient $3/\mathfrak{I}_3$ in $t_P = 3S_{\text{BH}}/\mathfrak{I}_3$ is purely algebraic: 3 is the entanglement normalisation ($c/6$ rather than $c/2$; equivalently, $\kappa/3$ rather than κ), and $\mathfrak{I}_3$ is the Virasoro complementarity constant $K = \mathfrak{I}_3$. The dependence on the specific black hole enters only through S_{BH} . For any Koszul pair with complementarity constant K , the Page time is $t_P = 3S_{\text{BH}}/K$.

9.3. The island as Koszul dual dominance. The island of the quantum extremal surface prescription has a precise algebraic incarnation: it is the locus in the genus expansion where the Koszul dual saddle dominates. The complementarity theorem fixes the number of saddles (exactly two), their curvatures (κ and κ'), and the transition time (t_P).

Theorem 9.7 (Complementarity decomposition). *At each genus $g \geq 1$, the genus- g obstruction decomposes under the Koszul involution:*

$$\mathbf{C}_g(\text{Vir}_c) = \mathbf{Q}_g(\text{Vir}_c) \oplus \mathbf{Q}_g(\text{Vir}_{26-c}).$$

The curvature of the first summand is $\kappa = c/2$; the curvature of the second is $\kappa' = (26 - c)/2$. There is no third saddle: the Koszul involution has period 2, so the pair $(\text{Vir}_c, \text{Vir}_{26-c})$ exhausts the decomposition.

The transition is sharp: at $t = t_P$, the dominance switches at every genus simultaneously because the ratio $\kappa/\kappa' = c/(26 - c)$ is genus-independent (the λ_g^{FP} factors cancel in the ratio of Boltzmann weights).

Proof. The Lagrangian eigenspace decomposition is Theorem C applied to the Virasoro algebra (Theorem 9.2). The curvatures κ and κ' are the eigenvalues of $d_{\text{fib}}^2/\omega_g$ on the two summands. The ratio of Boltzmann weights at genus g is

$$\frac{Z_+^{(g)}}{Z_-^{(g)}} = \exp((\kappa - \kappa') \lambda_g^{\text{FP}} \cdot 2^{g-2}) = \exp((c - \mathfrak{I}_3) \lambda_g^{\text{FP}} \cdot 2^{g-2}),$$

which depends on g through the common factor $\lambda_g^{\text{FP}} \cdot 2^{g-2}$ but whose sign ($c > \mathfrak{I}_3$ vs $c < \mathfrak{I}_3$) is genus-independent. At $c = \mathfrak{I}_3$: the ratio is 1 at all genera and all times. $\square$

Remark 9.8 (Comparison with Penington and AEMM). The gravitational island of Penington [6] and Almheiri–Engelhardt–Marolf–Maxfield [7, 8] is the locus where the quantum extremal surface dominates the gravitational path integral. The Koszul dual saddle $\tilde{B}(\text{Vir}_{26-c})$ is the algebraic counterpart: the two-saddle structure, the exchange of dominance, and the c -independent Page time all emerge from the complementarity constant $K = \mathfrak{I}_3$ alone. The identification:

QES / replica wormhole	Bar complex
no-island saddle	$\mathbf{Q}_g(\text{Vir}_c)$
island saddle	$\mathbf{Q}_g(\text{Vir}_{26-c})$
quantum extremal surface	Koszul involution fixed locus
island region	κ' -eigenspace dominance
Page time	saddle crossing
two saddles, no more	period-2 involution

The number of saddles (exactly two) is determined by the Koszul involution, not by a choice of replica geometry. The bar complex provides the mechanism; the Penington–AEMM programme provides the physical context and the fine-grained interpretation.

9.4. The transgression algebra and the Page transition. The transgression algebra B_Θ provides a second, independent characterisation of the Page transition. The degeneration of B_Θ at $u = \eta^2 = 0$ is the algebraic mechanism underlying the saddle exchange: the Morita-invisible Hawking phase degenerates into the information-carrying island phase.

Proposition 9.9 (The transgression algebra at the BTZ saddle). *Let B_Θ be the transgression algebra of the bar complex $\bar{B}(\text{Vir}_c)$ with MC element $\Theta = \Theta_{\text{Vir}_c}$ (Section 4). Set $u := \eta^2$ where η is the degree-one generator with $d\eta = \Theta$. At the BTZ saddle:*

- (i) *For $u \neq 0$ (the generic locus): the genus- g Clifford completion satisfies $\mathfrak{G}_g(B_\Theta)[u^{-1}] \cong \text{Mat}_{2^g}(B_\Theta[u^{-1}])$. Higher genus is Morita-invisible: the matrix algebra carries no new invariants beyond the genus-0 data. This is the **Hawking phase**: the boundary saddle dominates, and the black hole appears thermal.*
- (ii) *For $u = 0$ (the degenerate locus): $\mathfrak{G}_g(B_\Theta)/(\eta^2) \cong B/(\Theta) \otimes \Lambda^{2^g+1}$, the exterior algebra degeneration. The $\binom{2^g+1}{k}$ generators in Λ^k resolve the Morita-invisible matrix algebra into topological sectors of the genus- g partition function. This is the **island phase**: the Koszul dual saddle dominates, and the radiation carries fine-grained information.*
- (iii) *The **Page transition** is the passage from $u \neq 0$ to $u = 0$. The secondary anomaly $u = \eta^2$ vanishes when the two Lagrangian saddles reach equal weight. The locus $u = 0$ is the codimension-one wall in the space of MC deformations where the matrix algebra degenerates to the exterior algebra.*

Proof. Parts (i) and (ii) are the two regimes of the Clifford completion. The Clifford algebra $\text{Cl}_{2^g+1}(u)$ is the matrix algebra Mat_{2^g} for $u \neq 0$ (non-degenerate quadratic form) and the exterior algebra Λ^{2^g+1} for $u = 0$ (zero form).

Part (iii): the BTZ MC element $\alpha_{\text{BTZ}} = h e_h$ determines the curvature $\Theta|_\alpha = \kappa \cdot \omega_1$ at genus 1. As the black hole evaporates, $h(t) \rightarrow 0$ and $\alpha_{\text{BTZ}}(t) \rightarrow 0$; correspondingly, $u(t) = \eta^2|_{\alpha(t)} \rightarrow 0$. The Page time t_P is the time at which $u(t_P) = 0$ to leading semiclassical order, which coincides with the saddle crossing (15) because $u = 0$ is the locus where neither Lagrangian half dominates. $\square$

Remark 9.10 (Morita invisibility is thermality). The Morita equivalence $\mathfrak{G}_g \cong \text{Mat}_{2^g}(B_\Theta[\eta^{-2}])$ for $u \neq 0$ is the algebraic expression of Hawking’s argument: a thermal state carries no fine-grained information regardless of the number of replicas (g). The breakdown at $u = 0$ is the algebraic expression of unitarity restoration: the exterior algebra Λ^{2^g+1} resolves the thermal degeneracy. The Page curve interpolates between these two regimes; the passage is controlled by the scalar complementarity $\varkappa := \kappa + \kappa' = 13$ (the Trinity $K(\text{Vir}) = 26$; $\varkappa = K/2$).

9.5. The Borel singularity and Stokes phenomenon. The shadow free energies $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ converge (Bernoulli decay: $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$). The full gravitational genus expansion, including the field-sector

contributions from the infinite $\mathcal{A}_\infty$ tower (class **M**), diverges: the field-dependent terms $m_k|_T$ grow as $C^k \cdot k!$ (Gevrey-1, Remark 4.7). The Page transition has a third incarnation in this divergent structure: the Stokes phenomenon of the Borel-summed genus expansion.

Theorem 9.11 (Borel singularity from the Koszul dual). *The full gravitational genus expansion*

$$Z_{\text{grav}}(\cdot) = \exp\left(\sum_{g \geq 0} F_g^{\text{full}} \cdot 2^{g-2}\right),$$

where F_g^{full} includes both the convergent scalar shadow $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (UNIFORM-WEIGHT) and the Gevrey-1 field-sector corrections, satisfies:

- (i) The series diverges factorially: $|F_g^{\text{full}}| \sim C^g \cdot (2g)!$, forced by class **M** (all $m_k \neq 0$).
- (ii) The Borel transform $\hat{Z}(\zeta) = \sum_{g \geq 0} F_g^{\text{full}} \zeta^{2g-2} / (2g-2)!$ has a singularity at the instanton action

$$A_{\text{inst}} = \frac{|\kappa' - \kappa|}{K} \cdot (2\pi)^2 = \frac{|13 - c|}{13} \cdot 4\pi^2.$$

The singularity corresponds to the Koszul dual saddle $\bar{B}(\text{Vir}_{26-c})$.

- (iii) At $c = 13$ (self-dual): $A_{\text{inst}} = 0$. The Borel singularity sits at the origin, the two saddles have equal action, and the Stokes phenomenon is maximally strong.
- (iv) The Stokes wall of the Borel summation is the Page time: the exponentially suppressed correction $\sim e^{-A_{\text{inst}}/t}$ becomes of order unity at $t = t_P$ and dominates thereafter.

Proof. Part (i): the shadow depth $r_{\text{max}} = \infty$ (all $m_k \neq 0$, Theorem 2.10) forces the field-sector contributions to grow as $\|m_k|_T\| \sim C^k \cdot k!$ by the Virasoro OPE recursion. The cumulative genus- g free energy includes the sum over stable graph topologies at genus g with vertices decorated by $\{m_k\}$; the factorial growth of m_k combined with the graph-counting factor produces $|F_g^{\text{full}}| \sim C^g \cdot (2g)!$ (Gevrey-1). The scalar shadow $\kappa \cdot \lambda_g^{\text{FP}} \sim \kappa \cdot (2\pi)^{-2g}$ converges (Bernoulli decay) but the field sector dominates for $g \gg 1$.

Part (ii): in the resurgence framework, the Borel singularity nearest to the origin corresponds to the non-perturbative saddle closest in action. Complementarity provides exactly one such saddle: $\bar{B}(\text{Vir}_{26-c})$ with curvature κ' . The tree-level free energies are $\mathcal{F}_+^{(0)} = \kappa^{-2}$ and $\mathcal{F}_-^{(0)} = \kappa'^{-2}$; the action difference is $|\mathcal{F}_+^{(0)} - \mathcal{F}_-^{(0)}| = |\kappa - \kappa'|^{-2} = |c - 13|^{-2}/1$. Normalising by the scalar instanton action $(2\pi)^2$ (the leading pole of the $\hat{A}$ -genus at $x = 2\pi i$) gives $A_{\text{inst}} = (2\pi)^2 |\kappa' - \kappa|/K$.

Part (iii): at $c = 13$, $\kappa = \kappa'$, so $A_{\text{inst}} = 0$.

Part (iv): the Stokes wall is crossed when the Borel contour passes through the singularity. On one side: $\bar{B}(\text{Vir}_c)$ dominates. On the other: $\bar{B}(\text{Vir}_{26-c})$ dominates. The Stokes multiplier is ± 1 (the involution has period 2). The wall is crossed when the evaporating black hole reaches the critical value $t^{-1} = S_{\text{BH}}(t_P)$; at leading order this reproduces $t_P = 3S_0/13$. $\square$

Remark 9.12 (Three incarnations of the Page transition). The Page transition has three equivalent descriptions, each originating from a different structure on the bar complex:

- (1) **Lagrangian**: the crossing of the two Koszul-dual saddle-point contributions to the gravitational partition function (Theorem 9.4).
- (2) **Transgression**: the degeneration $u = \eta^2 \rightarrow 0$ in the transgression algebra B_Θ , where the Morita-invisible matrix algebra degenerates to the exterior algebra (Proposition 9.9).
- (3) **Resurgent**: the Stokes wall of the Borel-summed genus expansion, where the exponentially suppressed Koszul dual contribution becomes dominant (Theorem 9.11).

All three are controlled by the single datum $\varkappa := \kappa + \kappa' = 13$ (the scalar complementarity; the Trinity $K(\text{Vir}) = c + c^\dagger = 26$ sits upstream through $\varkappa = \varrho_{\text{Vir}} \cdot K = K/2$): it determines the crossing time, the degeneration locus, and the instanton action. The number of saddles is fixed by complementarity (period-2 involution), the degeneration type by the Clifford structure (matrix vs. exterior), and the Stokes multiplier is ± 1 (the involution sign).

10. DE SITTER ENTROPY AND THE GRAVITATIONAL YANGIAN

10.1. Analytic continuation to de Sitter. The algebraic Page curve exploits the Koszul pair $(\text{Vir}_c, \text{Vir}_{26-c})$ at fixed signature. Analytic continuation to Lorentzian de Sitter signature ($\ell \mapsto i\ell$, equivalently $\Lambda = -1/\ell^2 \mapsto +1/\ell^2$) preserves the shadow obstruction tower and the complementarity structure.

Theorem 10.1 (De Sitter shadow tower). *Under the Wick rotation $\ell \mapsto i\ell$ mapping $\text{AdS}_3 \rightarrow \text{dS}_3$:*

- (i) *The central charge continues to $c_{\text{dS}} = 3\ell_{\text{dS}}/(2G)$, and the curvature to $\kappa_{\text{dS}} = c_{\text{dS}}/2$.*
- (ii) *The Gibbons–Hawking entropy of the cosmological horizon is:*

$$S_{\text{dS}} = \frac{\pi c_{\text{dS}}}{3} = \frac{\pi \ell_{\text{dS}}}{2G} = \frac{\text{Area}(\text{cosmological horizon})}{4G}.$$

- (iii) *The shadow obstruction tower continues with real quantum corrections:*

$$F_g^{\text{dS}} = \kappa_{\text{dS}} \cdot \lambda_g^{\text{FP}} = \frac{c_{\text{dS}}}{2} \cdot \lambda_g^{\text{FP}} \quad (\text{UNIFORM-WEIGHT}).$$

In particular: $F_1^{\text{dS}} = c_{\text{dS}}/48$.

- (iv) *The Nariai limit $c_{\text{dS}} = 13$ is the analytic continuation of the self-dual point: the de Sitter cosmological horizon entropy at the Nariai bound is self-dual under $c_{\text{dS}} \mapsto 26 - c_{\text{dS}}$.*

Proof. The Brown–Henneaux formula $c = 3\ell/(2G)$ is algebraic in ℓ and continues under $\ell \mapsto i\ell$ (or, equivalently, $\ell_{\text{dS}} = |\ell|$). The curvature $\kappa = c/2$ is polynomial in c and continues directly. The Gibbons–Hawking entropy: the cosmological horizon of dS_3 has circumference $2\pi\ell_{\text{dS}}$, so $\text{Area} = 2\pi\ell_{\text{dS}}$, and $S_{\text{dS}} = 2\pi\ell_{\text{dS}}/(4G) = \pi\ell_{\text{dS}}/(2G) = \pi c_{\text{dS}}/3$. The shadow free energies $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ are polynomial in κ and continue to real values. The Nariai limit corresponds to the maximal black hole in dS_3 , where the black hole and cosmological horizons coincide; the central charge at this point is $c_{\text{dS}} = 13$, the self-dual value. $\square$

Remark 10.2 (The Banks conjecture). The Banks conjecture states that the Hilbert space of de Sitter quantum gravity is finite-dimensional: $\dim(\mathcal{H}_{\text{dS}}) = \exp(S_{\text{dS}}) = \exp(\pi c_{\text{dS}}/3)$. The convergent shadow obstruction tower (Bernoulli decay: $\lambda_g^{\text{FP}} \sim (2\pi)^{-2g}$) is consistent with this finite-dimensionality: the quantum corrections to the de Sitter entropy do not produce a divergent series that would require non-perturbative completion from additional degrees of freedom.

10.2. Horizon thermodynamics from negative curvature. The analytic continuation $\kappa \rightarrow -|\kappa|$ has a direct interpretation in the bar complex.

Proposition 10.3 (Negative curvature and horizon thermodynamics). *The fiberwise differential at genus $g \geq 1$ satisfies $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ on $\overline{\mathcal{M}}_g$. Under $\kappa \rightarrow -|\kappa|$:*

- (i) *The curved bar complex acquires opposite curvature: $d_{\text{fib}}^2 = -|\kappa| \cdot \omega_g$.*
- (ii) *The genus tower changes sign: $F_g^{\text{dS}} = -|\kappa| \cdot \lambda_g^{\text{FP}} < 0$ for $g \geq 1$.*
- (iii) *The negative free energies produce positive entropy contributions: $-F_g < 0$ in the exponent of Z means the partition function decreases, consistent with a finite-dimensional Hilbert space.*

The de Sitter entropy $S_{\text{dS}} = \pi c_{\text{dS}}/3$ is the genus-0 contribution (the classical action evaluated on the cosmological horizon saddle), with the tower of negative corrections $\{F_g^{\text{dS}}\}_{g \geq 1}$ systematically reducing the effective number of microstates.

Proof. The fiberwise differential is $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ (proved in Section 4). Replacing κ by $-|\kappa|$ changes the sign of ω_g . The genus- g free energy $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ then becomes $F_g^{\text{dS}} = -|\kappa| \cdot \lambda_g^{\text{FP}} < 0$ (since $\lambda_g^{\text{FP}} > 0$). $\square$

10.3. The gravitational Yangian at $c = 13$. At the Koszul self-dual point $c = 13$, the Virasoro algebra acquires a distinguished algebraic structure: the ordered bar complex $\bar{B}^{\text{ord}}(\text{Vir}_{13})$ carries a gravitational Yangian symmetry.

Definition 10.4 (Gravitational Yangian). The *gravitational Yangian* $\mathcal{Y}_{\text{grav}}$ is the algebra of conserved charges of the shadow obstruction tower $\{S_r\}_{r \geq 2}$ at $c = 13$, acting on the ordered bar complex $\bar{B}^{\text{ord}}(\text{Vir}_{13}) = T^c(\mathcal{S}^{-1}\bar{\text{Vir}}_{13})$. The shadow connection $\nabla^{\text{hol}} = d - \sum_{r \geq 2} S_r$ at $c = 13$ is the generating function of the Yangian generators.

Theorem 10.5 (Properties of the gravitational Yangian at $c = 13$). *At the self-dual point $c = 13$:*

- (i) **Self-duality.** $\text{Vir}_{13}^! = \text{Vir}_{13}$: the boundary algebra equals its Koszul dual. The curvature is $\kappa = 13/2$, and $\kappa_{\text{eff}} = (13 - 26)/2 = -13/2$.
- (ii) **Symmetric genus tower.** $F_g(13) = F_g(26 - 13) = F_g(13)$: the genus tower is self-dual. The free energies: $F_1 = 13/48$, $F_2 = 91/11520$.
- (iii) **Symmetric Page curve.** At $c = 13$: $\kappa = \kappa' = 13/2$, the two saddles have equal action, the Page curve is symmetric about t_P , and $S_{\text{Page}} = S_{\text{BH}}/2$.
- (iv) **Infinite conserved charges.** The shadow obstructions S_r , $r \geq 2$, at $c = 13$ satisfy:

$$\begin{aligned} S_2 &= 13/2, & S_3 &= -13, & S_4 &= 10/(13 \cdot 87) = 10/1131, \\ S_5 &= -48/(169 \cdot 87) = -48/14703. \end{aligned}$$

The discriminant $\Delta = 8 \cdot (13/2) \cdot 10/1131 = 40/87 \neq 0$ confirms class **M**: the gravitational Yangian has infinitely many generators.

- (v) **R-matrix at self-duality.** The classical r -matrix at $c = 13$ is $r_{13}(z) = (13/2)/z^3 + 2T/z$, and the CYBE holds by Proposition 2.7. The KZ connection (13) at $c = 13$ has self-dual monodromy: the braiding σ satisfies $\sigma^2 = \sigma_{\text{dual}}^2$ (the braiding of Vir_{13} and $\text{Vir}_{13}^! = \text{Vir}_{13}$ coincide).

Proof. Parts (i)–(iii) are immediate from Theorem 5.1 and Proposition 5.2 at $c = 13$.

Part (iv): the shadow invariants at $c = 13$ are obtained by evaluating the general formulas: $S_2 = 13/2$, $S_3 = -13$, $S_4 = 10/(13(5 \cdot 13 + 22)) = 10/(13 \cdot 87) = 10/1131$, $S_5 = -48/(13^2 \cdot 87) = -48/14703$. The discriminant $\Delta = 8\kappa S_4 = 8(13/2)(10/1131) = 40/87 \neq 0$; class **M**.

Part (v): the r -matrix at $c = 13$ is $r_{13}(z) = (13/2)/z^3 + 2T/z$ by direct evaluation of (8). The CYBE holds for all c by Proposition 2.7. Koszul self-duality of the monodromy at $c = 13$ (i.e. $\text{Vir}_{13}^! = \text{Vir}_{13}$, the Koszul involution $c \mapsto 26 - c$ fixing $c = 13$) implies that the Koszul dual connection is the same connection. $\square$

Remark 10.6 (The role of $c = 13$ versus $c = 26$). The two distinguished central charges are:

- $c = 13$: *Koszul self-duality.* $\text{Vir}_{13}^! = \text{Vir}_{13}$. The complementarity constant is $K = 13$. The algebraic Page curve is symmetric. The gravitational Yangian has full self-dual structure. This is the algebraic self-duality of 3d gravity.

- $c = 26$: *matter-ghost cancellation*. $\kappa_{\text{eff}} = 0$. The effective genus tower vanishes: $F_g^{\text{eff}} = 0$ for all $g \geq 1$. The gravitational partition function becomes S -invariant. This is the bosonic string critical point, not self-duality.

The physical content of 3d gravity concentrates at $c = 13$, not $c = 26$: at $c = 13$, the gravitational perturbation series (the shadow obstruction tower) has maximal symmetry (self-duality), the algebraic Page curve has maximal symmetry (equal entropies), and the Yangian has maximal structure (self-dual braiding). At $c = 26$, the effective perturbation series vanishes.

Conjecture 10.7 (The gravitational Yangian as symmetry algebra). *At $c = 13$, the gravitational Yangian $\mathcal{Y}_{\text{grav}}$ is the symmetry algebra of 3d self-dual quantum gravity in the following sense:*

- (i) *The shadow obstructions $\{S_r\}_{r \geq 2}$ generate an algebra isomorphic to a quotient of the Yangian $Y(\mathfrak{vir}_{13})$ associated to the Virasoro Lie algebra at $c = 13$.*
- (ii) *The shadow connection $\nabla^{\text{hol}} = d - \sum_{r \geq 2} S_r$ is a connection for $\mathcal{Y}_{\text{grav}}$ -modules.*
- (iii) *The integrability of the self-dual gravitational theory (manifested by the convergence of the scalar shadow tower) is a consequence of the Yangian symmetry.*
- (iv) *The conserved charges S_r at $c = 13$ satisfy the RTT-type commutation relations*

$$R_{12}(z - w) T_1(z) T_2(w) = T_2(w) T_1(z) R_{12}(z - w)$$

with R -matrix constructed from $r_{13}(z)$.

Remark 10.8 (Comparison with the holographic datum). The gravitational Yangian at $c = 13$ is the most structured specialisation of the holographic datum $\mathcal{H}(\text{Vir}_c)$: at $c = 13$, all six components are Koszul self-dual (invariant under $c \mapsto 26 - c$), the algebraic Page curve is symmetric, the genus tower has maximal symmetry, and the Yangian structure organises the infinite $\mathcal{A}_\infty$ tower into a representation-theoretic framework. The holographic datum at generic c carries the same six components but without the Koszul self-duality (specific to $c = 13$) that enables the Yangian interpretation.

II. WAVE-14 RECONCEPTION: GRAVITY AS PLATONIC CLIMAX

Three-dimensional quantum gravity is not a motivational appendix to the modular Koszul programme; it is the platonic climax. This section sharpens the relationship between the body of the paper and the 2026-04-17 platonic-ideal anchors: the iterated Sugawara E_∞ -topologisation ladder, the chiral higher-Deligne identification of the bulk, universal celestial holography, and the topologization tower across all four shadow classes.

Theorem 11.1 (Iterated Sugawara E_∞ -topologisation ladder). *For a chiral algebra $\mathcal{A}$ with a tower of inner higher-spin Casimirs $\{T^{(n)}\}_{2 \leq n \leq N+1}$, each admitting a BRST primitive $G^{(n)}$ with $T^{(n)} = [Q_{\text{tot}}, G^{(n)}]$ on Q -cohomology, the derived chiral centre carries an E_{N+2} -topological structure via iterated Dunn additivity. Specialisations:*

- $\mathcal{A} = \text{Vir}_c$: one inner stress tensor, E_3^{top} (this paper's Theorem 3.3);
- $\mathcal{A} = \mathcal{W}_N$: inner tower of depth $N - 1$, E_{N+1}^{top} ;
- $\mathcal{A} = \mathcal{W}_\infty$: infinite depth, E_∞^{top} (platonic endpoint).

Consequently, 3d quantum gravity (the Vol II gravitational climax, Part VI of that volume) is the $N = 2$ shadow of a $3d + \infty$ topological theory.

Theorem 11.2 (Chiral higher Deligne and universal holography). *The derived chiral centre $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$ is the universal E_3 -chiral algebra acted on by $\text{SC}^{\text{ch}, \text{top}}$ through the heptagon edges (3) $\leftrightarrow$ (4) $\leftrightarrow$ (5)*

of the Vol II $SC^{\text{ch},\text{top}}$ heptagon. On the C_2 -cofinite non-logarithmic standard landscape, this produces universal celestial holography: the celestial chiral algebra $\mathcal{A}^{\text{cel}}$ of boundary soft modes, the $SC^{\text{ch},\text{top}}$ structure on $(\mathcal{A}^{\text{cel}}, Z_{\text{ch}}^{\text{der}}(\mathcal{A}^{\text{cel}}))$, and the shadow-tower coefficients reproducing the soft-factor hierarchy. Coverage: self-dual gauge (KM), gauge+matter (DS), gravity (Virasoro + $w_{1+\infty}$).

Remark 11.3 (Topologization tower: scope per class). Topologization of $Z_{\text{ch}}^{\text{der}}(\mathcal{A})$ has four regimes, with scope tags attached for every reference in this paper:

- Class G (Heisenberg): E_3^{top} immediate via Dunn on F^{H_k} (commutative $\Rightarrow$ Dunn applies).
- Class L ($V_k(\mathfrak{g})$, $k \neq -h^\vee$): E_3^{top} on the *original* complex via Costello–Gwilliam factorization on $\mathbb{R} \times \mathbb{C}$ with explicit topologizing primitives $\eta_i^{(i)}$, $\eta_i^{(ii)}$ making $[Q, \tilde{G}_1] = T_{\text{Sug}}$ strict chain-level.
- Class C ($\beta\gamma$, bc): E_3^{top} via Friedan–Martinec–Shenker bosonisation, reducing to Heisenberg.
- Class M (Vir, $\mathcal{W}_N$): E_3^{top} weight-completed via half-BRST + MC₅ pattern; original-complex chain-level *open* (sole remaining direction of the topologization tower theorem).
- Critical level $k = -h^\vee$: E_2^{top} (dimension drop, not failure).

The present paper’s Theorem 3.3 is the class M line of this tower, attached with the cohomological / weight-completed scope tag (AP-TOPOLOGIZATION).

Remark 11.4 (Programme status: the open frontier, pre/post 2026-04-17). The only remaining open frontier of the 2026-04-16 closure wave is class M chain-level topologization on the *original* complex; it was closed weight-completed, not chain-level. The 2026-04-17 Beilinson audit added the following scope qualifications: (i) $W(p)$ triplet tempering is conjectural pending an Adamović–Milas character-amplitude bound (Gurarie 1993 + Flohr 1996 counterexamples); (ii) non-tempered stratum is *empty* only on the non-logarithmic subset; (iii) the β_N closed form differs at $N = 4$ (15 vs 16), parameter-question not structural; (iv) super-Yangian canonical pairing is sub-Sugawara. 3d quantum gravity with $\Lambda < 0$, the Virasoro climax at $c = 3\ell/(2G)$, and the c -linearity/self-duality structure are unconditional on the Virasoro Koszul locus (generic c); the class M chain-level conditional affects only original-complex chain-level bar/BV comparison, not the genus expansion computed in Section 4.

Remark 11.5 (The climax is the theorem, not the motivation). The relationship “3d quantum gravity IS the platonic theorem” is precise: the universal Maurer–Cartan element Θ_{Vir_c} of the modular convolution algebra encodes, via its MC projections to successive degrees, the complete perturbative gravitational genus expansion. Koszul duality $\text{Vir}_c \leftrightarrow \text{Vir}_{26-c}$ is gravitational S-duality; Page-time complementarity $t_P = 3S_{\text{BH}}/13$ is the structural shadow of the Page transition (the full information-theoretic content requires the island mechanism, which is supplied by the bulk E_3 -topological structure of $Z_{\text{ch}}^{\text{der}}(\text{Vir}_c)$). BTZ black holes are MC elements; de Sitter entropy emerges from analytic continuation $\kappa \rightarrow -|\kappa|$. The iterated Sugawara ladder (Theorem 11.1) reveals this gravity as the $N = 2$ *shadow* of an $N \rightarrow \infty$ topological tower: the full platonic endpoint is E_∞^{top} gravity in $3 + \infty$ dimensions, of which 3d quantum gravity is the lowest truncation.

REFERENCES

- [1] A. Achúcarro and P. K. Townsend, *A Chern–Simons action for three-dimensional anti-de Sitter supergravity theories*, Phys. Lett. B **180** (1986), 89–92.
- [2] E. Witten, *(2 + 1)-dimensional gravity as an exactly soluble system*, Nucl. Phys. B **311** (1988), 46–78.
- [3] J. D. Brown and M. Henneaux, *Central charges in the canonical realization of asymptotic symmetries: an example from three-dimensional gravity*, Comm. Math. Phys. **104** (1986), 207–226.
- [4] M. Bañados, C. Teitelboim, and J. Zanelli, *The black hole in three-dimensional space-times*, Phys. Rev. Lett. **69** (1992), 1849–1851.

- [5] B. Ponsot and J. Teschner, *Clebsch–Gordan and Racah–Wigner coefficients for a continuous series of representations of $U_q(\mathfrak{sl}(2, \mathbb{R}))$* , Comm. Math. Phys. **224** (2001), 613–655.
- [6] G. Penington, *Entanglement wedge reconstruction and the information problem*, J. High Energy Phys. **2020** (2020), no. 9, 002; arXiv:1905.08255.
- [7] A. Almheiri, N. Engelhardt, D. Marolf, and H. Maxfield, *The entropy of bulk quantum fields and the entanglement wedge of an evaporating black hole*, J. High Energy Phys. **2019** (2019), no. 12, 063; arXiv:1905.08762.
- [8] A. Almheiri, R. Mahajan, J. Maldacena, and Y. Zhao, *The Page curve of Hawking radiation from semiclassical geometry*, J. High Energy Phys. **2020** (2020), no. 3, 149; arXiv:1908.10996.

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